

Variation and Evolution of Mammalian Corticospinal Somata With Special Reference to Primates

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ABSTRACT

The morphology of the somata originating the corticospinal tract was examined in 24 species of mammals to identify commonalities and major sources of variation among the different species. Horseradish peroxidase was applied to a hemisection of the spinal cord at the C1–C2 junction. After tetramethylbenzidine processing, the labeled somata throughout the cerebral cortex were plotted and counted. Then, 23 morphological characteristics of the corticospinal somata were examined, including their number, size, and density across the cortical surface.

The results show that morphological characteristics of corticospinal somata are closely related to an animal's body, brain, and cerebral cortex size. That is, mammals with large neocortical surfaces tend to have larger as well as more corticospinal somata; mammals with large bodies tend to have corticospinal somata that are less densely distributed. Moreover, the probable increase in the ratio of local noncorticospinal somata to corticospinal somata implies that the evolution of the corticospinal tract was accomplished by an increase in "support" or "server" cells as well as an increase in the size of the tract itself.

The results also show that several characteristics are reliably related to an animal's taxonomic classification and hence its ancestry. Comparisons among three mammalian lineages indicate that some characteristics may have changed uniquely in the anthropoid primate lineage, and thus, presumably, in the human lineage. The results suggest that if morphological characteristics of the corticospinal tract important in the evolution of the specialized motor abilities in anthropoid primates are sought, then examination of the role of changes in soma diameter, rostral (motor)/caudal (sensory) ratios of density, concentration, surface density, and volume density may be more instructive than examination of the total number of corticospinal neurons alone. © 1995 Wiley-Liss, Inc.

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The corticospinal tract (CST) places the cerebral neocortex in direct neural contact with the spinal cord. The presence of monosynaptic connections between cortical neurons and spinal motoneurons in primates has been the chief structural basis for implicating the CST in a variety of motor functions, such as the control of digital dexterity, conditioned movements, flexor activity, muscle tonus, and volitional movements (e.g., Bernhard and Bohm, 1954; Buxton and Goodman, 1967; Jane et al., 1968; Lawrence and Kuypers, 1968; Beck and Chambers, 1970; Heffner and Masterton 1975; Wiesendanger, 1981, 1984; Heffner and Masterton 1983; Alstermark et al., 1989; Davidoff, 1990; Nathan et al., 1990).

Because rather strong structure–function relationships for the CST can be advanced in primates, including humans, it has been tempting to presume that its function may be the same in all mammals. However, it is now

becoming apparent that because the morphology of the CST is so varied across mammals in size, spinal course, manner of termination, distribution of cells of origin, etc., most motor functions ascribed to the CST of primates probably cannot be extended to other orders of mammals or vice-versa.

Although the anatomy and physiology of the CST have been studied repeatedly in several mammalian species over the past century, comprehensive comparative investigations have been infrequent. In the few that have been done, conclusions about the tract's phylogenetic origins and

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subsequent evolution have been based almost exclusively on its course down the spinal cord and, to a lesser extent, on its terminal distributions (e.g., Goldby, 1939; Lassek, 1941; Nyberg-Hansen and Brodal, 1963; Bautista and Matzke, 1965; Campbell et al., 1966; Scheibel and Scheibel, 1966; Petras, 1968; Fisher et al., 1969; Harting and Noback, 1970; Martin et al., 1970, 1975; Goode and Haines, 1975; Heffner and Masterton 1975, 1983; Satomi et al., 1989; Wang et al., 1992). Over the past decade, however, with the help of reliable retrograde tract-tracing techniques, it has been possible to plot the cortical origins of the CST in any mammal. The resulting data can then be subjected to statistical analyses testing ideas surrounding the evolutionary history or modes of variation of the CST.

Because corticospinal (CS) fibers probably exist in all extant mammals, comparative methodology suggests that the CST must have emerged very early in the history of mammals, before the divergence of all extant mammals. This means that a reasonably dense sample of an evolutionary sequence of humankind's extinct ancestors can be approximated from the characters in common of selected extant mammals. By reviewing its changing form across mammals arranged according to their relative propinquity, or recency of common ancestry with anthropoid primates (and thus, presumably, with humans), it is hoped that further insight into the evolution of neocortex, its direct descending outflow, and its role in specific human functions can be gained.

Considerable comparative information already exists on the morphology of the fibers of the pyramidal tract. These studies demonstrate that the CST varies considerably among mammals in its origin, size, funicular trajectory, extension down the spinal cord, and laminar termination (e.g., Spitzka, 1886; Holmes and Page May, 1909; Lassek, 1941, 1948; Campbell, 1965; Verhaart, 1970; Phillips, 1971; Towe, 1973; Heffner and Masterton, 1975, 1983; Wiesendanger, 1981; Nudo and Masterton, 1990a). For example, Heffner and Masterton (1975, 1983) showed that in primate evolution, CS axons probably extended farther down the spinal cord and terminated progressively closer to motor neurons. In addition, the cross-sectional area of the pyramidal tract probably increased throughout primate evolution. However, relatively little comparative information has been available concerning the somata originating the CST. Often it has been stated that certain morphological characteristics of CS neurons are better developed in primates than in other mammalian species (e.g., Lassek, 1948). To the extent that morphological changes in CS neurons may parallel the evolutionary development of the specialized motor skills present in humans, it is useful to examine which, if any, characteristics may have changed uniquely in the primate ancestral lineage.

In previous papers, we described and analyzed the distribution of CS somata across the neocortical surface in nearly two dozen diverse mammalian species (Nudo and Masterton, 1990a,b). These studies demonstrated several characteristics of CS somata morphology common to each animal in the sample and, hence, probably common to most if not all mammals. For example, the tract originates from two spatially segregated regions of cortex but exclusively from layer V wherever its origin. We have called these separate areas "region A" (which is the sum of primary somatosensory and motor areas, or SI and MI, the supplementary motor area, or SMA, and posterior parietal cortex) and "region B" (which is approximately equivalent to the second somatosensory area, or SII). Furthermore, these

same comparative studies have shown that some characteristics of CS neurons are probably unique to specific subgroups of mammals. That is, in addition to these two regions, which are probably present in all mammals, a third CS region ("region C") is present in the cortex of primates and an apparently different third region ("region C'") is present in the cortex of Glires (Rodentia and Lagomorpha¹). Quantifying these results in terms of the area of the cortex devoted to the CST showed that the relative amount of such cortex probably remained constant over a very long period of anthropoid ancestry. Analyzing these same data in functional and ecological terms, the distribution of CS somata among the three cortical regions in primates was found to be more closely related to their particular mode of adaptation than to their particular combination of digital dexterity or hand-eye coordination.

In the present paper, we examine several other morphological characteristics—this time, the characteristics of the CS somata themselves. We have included the total numbers and densities in each of the regions, the soma diameters of CS neurons, and the distribution of CS somata within the principal region, region A. By examining the variation in form of the CS somata among mammals, we derive the evolutionary trends that seem to have been the most persistent and profound in mammalian evolution. Finally, we test the possibility that some of these trends might have been unique to the anthropoid ancestral lineage.

First, by using statistical tests of relationships to an animal's body size, brain size, or cerebral neocortex area, we examine the contribution of allometric (or scaling) factors to the variation in CS somata morphology. Second, we examine the contribution of an animal's heritage or ancestral lineage to the variation in CS somata morphology not related to size alone. These analyses allow us to deduce the most probable course of evolution of the anthropoid primate CST. In this way, we may determine which aspects of CS morphology are important in specialized motor skills of anthropoid primates and, conversely, determine which aspects of CS morphology probably make a lesser contribution to specialized skills in anthropoid primates.

MATERIALS AND METHODS

The selection of a retrograde tract-tracing procedure that would maximize the labeling of supraspinal cell groups, the experimental basis for the surgical and histological techniques eventually selected, and the screening of the histological materials prior to quantification have been described in detail elsewhere (Nudo and Masterton, 1988, 1989, 1990a,b, 1993). Briefly, after a preliminary series of experiments with rats using several retrograde tracers, sites and methods of application, survival times, and chromogens, the horseradish peroxidase (HRP) bulk-filling procedure of Hicks and D'Amato (1977) with tetramethylbenzidine processing (Mesulam, 1978) was adopted as a standardized procedure for comparative purposes. At least in our hands, this procedure was not surpassed by any other technique for labeling the largest numbers of supraspinal neurons in each species.

In our standard version of this procedure, the animal was weighed, then anesthetized with 3.3 cc/kg of a mixture of

¹There is still considerable disagreement among mammalogists concerning the validity of the cohort Glires (e.g., Novacek, 1992).

TABLE 1. Body Weight, Brain Weight, and Neocortical Surface Area for Species and Genera Represented

Common name	Scientific name	Body weight (g)	Brain weight (mg)	Cortical surface area (mm ²)
Albino rat	<i>Rattus norvegicus</i>	268	1,500	152
Nine-banded armadillo	<i>Dasypus novemcinctus</i>	5,000	7,540	308
Lesser bushbaby	<i>Galago senegalensis</i>	160	2,400	436
Domestic cat	<i>Felis catus</i>	3,068	19,080	2,301
Eastern gray squirrel	<i>Sciurus carolinensis</i>	400	4,930	388
13-lined ground squirrel	<i>Citellus tridecemlineatus</i>	150	1,530	188
West African hedgehog	<i>Erinaceus albiiventris</i>	410	1,300	81
European hedgehog	<i>E. europaeus</i>	315	2,260	193
Desert hedgehog	<i>Hemiechinus auritus</i>	203	965	119
Rock hyrax	<i>Procavia capensis</i>	2,500	12,420	836
Least shrew	<i>Cryptotis parva</i>	8	80	18
Common marmoset	<i>Callithrix jacchus</i>	287	5,480	892
Eastern American mole	<i>Scalopus aquaticus</i>	53	570	87
Green monkey	<i>Cercopithecus aethiops</i>	1,370	36,540	3,616
Crab-eating macaque	<i>Macaca fascicularis</i>	1,932	38,030	5,548
Rhesus macaque	<i>M. mulatta</i>	3,300	53,700	7,546
Squirrel monkey	<i>Saimiri sciureus</i>	691	17,040	2,150
North American opossum	<i>Didelphis virginiana</i>	2,727	4,360	316
Gray, short-tailed opossum	<i>Monodelphis domestica</i>	83	790	68
Rabbit	<i>Oryctolagus cuniculus</i>	5,230	7,570	634
Raccoon	<i>Procyon lotor</i>	4,091	31,870	3,669
Slow loris	<i>Nycticebus coucang</i>	965	7,880	837
Tree shrew	<i>Tupaia glis</i>	230	2,480	290
Pine vole	<i>Pitymys pinetorum</i>	16	370	55

chloral hydrate and pentobarbital (Chloropent, Fort Dodge). After the spinal cord was hemisected at the C1–C2 junction, flakes of raw HRP were applied directly to the freshly cut fibers. At the end of a 3-day survival period, the animal was injected with a lethal dose (10.0 cc/kg) of Chloropent and perfused. The brain and spinal cord were then removed and immersed in 30% sucrose at 4–10°C. After the tissue had sunk (1–2 days), it was weighed. Then it was cut in 50- μ m frontal or sagittal sections. The resulting materials were processed with tetramethylbenzidine. This procedure results in Golgi-like filling of CS somata, apical and basal dendrites, and axons. This procedure was applicable to each species, and therefore each animal was treated in a virtually identical way. After a rigorous screening of the histological materials to eliminate cases with either incomplete or misplaced hemisections, extracellular HRP in the dorsal column nuclei, or cell labeling indicative of HRP transport by blood or cerebrospinal fluid, the individual cases were admitted for quantification (Nudo and Masterton, 1988, 1989).

Species selection

The results from animals representing 24 species in 21 genera are reported here. These 21 genera, in turn, represent 16 families, 9 orders, and 2 subclasses of Mammalia (Table 1).

Several considerations guided our choice of these particular animals. To begin with, 14 of the species were selected on paleontological, taxonomic, and comparative anatomical grounds for their sequential kinship, or relative recency of common ancestry, with Old World monkeys. This selection allowed the representation of five grades of successive kinship with Old World monkeys: grade 1, marsupials;

grade 2, eutherians (nonprimates); grade 3, prosimian primates; grade 4, New World monkeys; and grade 5, Old World monkeys. However, the phylogenetic classification of New World monkeys is not yet settled, especially within the New World simian infraorder Platyrrhini, and alternative schemes may prove to be valid (e.g., see McKenna, 1975; Eisenberg, 1981; Martin, 1990; Novacek, 1992). Nevertheless, the five grades of successive kinship with Old World monkeys are used here because they are generally accepted based on available paleontological data, and because they have been used in prior studies using this same set of species (Nudo and Masterton, 1989, 1990b).

To these, armadillo was added because edentates may form a sister group of all other placental mammals (Novacek, 1992); tree shrew was added because of their close relationship to primates. In the present analysis, however, we have included armadillo and tree shrew with the insectivores to (and hence with other Eutherian nonprimates) maintain the more conservative five grades of successive kinship.

To determine whether or not trends in the evolution of the CST in the Old World monkey lineage are unique to that lineage, several other species were selected to approximate sequential kinship with two additional mammalian lineages. Two species of Carnivora were selected (cat, raccoon) on the basis of their successive kinship with raccoons, a lineage superficially paralleling the origin of primate manual dexterity. Of the remaining six species, five were selected from the orders of Rodentia and Lagomorpha (rabbit, rat, vole, ground squirrel, tree squirrel) on the basis of their successive kinship with tree squirrels, a lineage superficially paralleling the origin of a primate visual system.

Finally, we added rock hyrax (*Procavia capensis*) to that list, mostly because it seems to illustrate an ipsilateral CS system and because of its use in previous comparative or phylogenetic arguments (Verhaart, 1967), bringing the total species to 24.

Quantification of results

To obtain a reasonable estimate of the total number of CS neurons in each animal, the HRP-labeled profiles were counted in a series of at least every fifth 50- μ m section throughout the rostral-caudal length of the cerebral hemisphere (or medial-lateral length of parasagittally sectioned brains) contralateral to the spinal cord hemisection. The raw counts of labeled profiles in the cerebral hemisphere were subjected to appropriate stereological corrections for soma size, section thickness, and minimum countable profile size (e.g., Weibel, 1979). These corrected cell counts were then systematically compared among the species for their possible relationship to the body, brain, and neocortical size of their hosts.

Cresyl-violet counterstained sections were used for measurements of the thickness of the layer of labeled somata within cortical layer V and for the concentration of labeled somata (discussed later).

The measurements of body weight, brain weight, and neocortical surface area have been described elsewhere (Nudo and Masterton, 1989, 1990b). They are retabulated here in Table 1 for easy reference. To measure the total neocortical surface area, we used a computerized planimeter on optical projections of systematic sets of coronal or sagittal sections that had been cresyl-violet counterstained. For each section, we measured the dorsal-ventral length of

the entire neocortical surface from its border with the cingulate cortex to the depths of the rhinal fissure (in coronally sectioned brains) or from its rostral to its caudal limits (in sagittally sectioned brains). By multiplying the distance between the sections by the sum of each set of planimetric measurements, we obtained estimates of the areas of the entire neocortical surface. The measurement error inherent in this technique is less than 3% for the total neocortical area (Nudo and Masterton, 1990b).

The total of 23 characteristics of CS somata obtained are divided into two broad classes: direct measurements, i.e., those characteristics that were directly measured from the histological materials (e.g., total numbers of CS somata, CS soma diameter, column height, etc.); and derived measurements, i.e., secondary results derived from the measurements by arithmetic operations (e.g., surface density, volume density, concentration, etc.).

Direct measurements

Total number of labeled somata. The total number of labeled somata used in the present analysis is the stereologically corrected total number in the individual with the most labeled somata, i.e., the maximum number found among any of the several representatives of that species. These values should most closely approximate the true number of CS somata because of the elimination of cases with possibly incomplete hemisections or incomplete labeling (see Nudo and Masterton, 1989). When identical analyses were performed using species means instead of maxima, the results were not significantly different.

The total number of labeled somata used in statistical analyses is the total in the neocortex contralateral to the spinal cord hemisection in each species except the three hedgehogs, mole, and hyrax. In these species, the total in the ipsilateral neocortex was used because the CST is predominantly uncrossed, at least at its entry into the cord, and hence the total in the ipsilateral hemisphere always exceeded the total in the contralateral hemisphere (Nudo and Masterton, 1990a).

Total number of labeled somata in each region. As discussed earlier, the CST in mammals originates either in two or in three separate regions of neocortex. Region A includes the frontal and parietal cortex, roughly corresponding to primary motor cortex (MI), premotor cortex, supplementary motor area (SMA), primary somatosensory cortex (SI), and posterior parietal cortex; region B corresponds roughly to the second somatosensory area (SII); region C corresponds to the arcuate (or lateral) premotor area (APA); and region C' of rodents and rabbit corresponds to the rostral forelimb area (Neafsey et al., 1986; Nudo and Masterton, 1990a,b; Rouiller et al., 1993).

Soma diameter. The soma size measurements used both as raw data and for the stereological corrections for overcounting were measured with the aid of a microscope and computerized planograph. Within each brain, individual somata were chosen randomly from the cortical areas containing the highest density of labeled somata within each brain. Only neurons whose apical dendrite was entirely within the plane of section were selected for the soma-size measurement. The perimeter of each soma was then measured and converted into an equivalent diameter.

Maximum thickness of the layer of labeled somata. As reported previously, we found no exception to the rule that CS somata are invariably contained within layer V and only layer V in each species examined (Nudo and Masterton,

1990a). However, the thickness of the layer of CS somata does vary among mammals and appeared worthy of measurement. The thickness of the layer of labeled somata, hereafter called "thickness," is the maximum distance measured perpendicular to the cortical surface from the dorsalmost to the ventralmost labeled neuronal somata within a small patch of layer V.

Maximum column height of labeled somata within layer V. Because labeled somata often appeared to be vertically arranged, one on top of another in the same focal plane within layer V, and because the maximum number of labeled somata within the column was variable among the animals, a measure of this characteristic was also made. Maximum column height is the maximum number of labeled somata (which occur above and below one another) within layer V and within a $30\ \mu\text{m} \times 50\ \mu\text{m}$ wide slab perpendicular to the cortical surface.

Medial-lateral position of zone of maximum density within region A. The medial-lateral position of the zone of maximum density within region A varied among the species. To quantify this characteristic, we used the percentage distance of this peak density from the most medial to the most lateral labeled neuron.

Derived measurements

Given the six direct measurements described earlier, seven other useful descriptions of the variation among species could be derived from their combination.

Percentage of labeled somata by region. The percentage of labeled somata in each region is the total number of labeled somata for a given region divided by the total number of labeled somata in all regions.

Maximum surface density of labeled somata by region. The maximum surface density is the maximum number of labeled somata under each unit surface area of neocortex. These surface densities of labeled cells were estimated by counting the number of labeled somata subjacent to each $250\ \mu\text{m} \times 50\ \mu\text{m}$ area of neocortical surface within each section.

Average surface density. The average surface density is the total number of labeled somata in neocortex divided by the total neocortical surface area containing the labeled somata.

Maximum volume density of labeled somata. Because the thickness of layer V varies across the cortex and among species, maximum volume density was also measured. Maximum volume density is the maximum number of labeled somata per unit volume of the labeled layer V of cortex. Arithmetically, the maximum volume density is the maximum surface density divided by the thickness of the layer of labeled somata.

Maximum concentration of labeled layer V somata and concentration of large layer V somata. The concentration of labeled somata is the number of labeled somata divided by the total number of somata within layer V subjacent to a $1\ \text{mm} \times 50\ \mu\text{m}$ sample area of cortex. In other words, concentration is equivalent to the proportion of CS somata within layer V. To derive the maximum concentration, the zone of maximum surface density was chosen for the sample. Then, a $1\ \text{mm} \times 50\ \mu\text{m}$ area was divided into four $250\text{-}\mu\text{m}$ samples. The maximum percentage among the four samples was defined as the maximum concentration. Because unlabeled somata approaching the size of the largest labeled somata were only rarely observed, a second measure of concentration was also made in which only those labeled

TABLE 2. Number and Percentage of Corticospinal Somata in Each Region

Animal/G.s. ¹ /Order	Maximum profile # ²	Soma diameter (μm)	Correction term	Corrected #CSN	Region								# Other	% Other
					#A	%A	#B	%B	#C	%C	#C'	%C'		
Albino rat/R.n./Rodentia	21,825	19.52	0.837	18,268	15,363	84.1	749	4.1	0	0.0	2,119	11.6	37	0.2
Armadillo/D.n./Edentata	12,505	19.57	0.836	10,454	10,088	96.5	366	3.5	0	0.0	0	0.0	0	0.0
Bushbaby/G.s./Primates	45,335	18.33	0.845	38,308	34,516	90.1	3,294	8.6	421	1.1	0	0.0	77	0.2
Cat/F.c./Camivora	41,820	26.61	0.790	33,038	29,866	90.4	2,412	7.3	0	0.0	0	0.0	760	2.3
Gray squirrel/S.c./Rodentia	19,695	21.89	0.820	16,150	13,308	82.4	1,308	8.1	0	0.0	1,486	9.2	48	0.3
Ground squirrel/C.t./Rodentia	24,775	18.30	0.845	20,935	18,695	89.3	1,151	5.5	0	0.0	1,089	5.2	0	0.0
Hedgehog/E.a./Insectivora	765 (2,135)	12.57	0.888	1,896	1,644	86.7	252	13.3	0	0.0	0	0.0	0	0.0
Hedgehog/E.e./Insectivora	4,255 (5,545)	19.42	0.837	4,641	3,550	76.5	1,091	23.5	0	0.0	0	0.0	0	0.0
Hedgehog/H.a./Insectivora	615 [1,415]	16.76	0.856	1,211	957	79.0	254	21.0	0	0.0	0	0.0	0	0.0
Hyrax/P.c./Hyracoidea	5,990 (25,895)	25.42	0.797	20,638	19,008	92.1	1,548	7.5	0	0.0	0	0.0	82	0.4
Least shrew/C.p./Insectivora	474	12.08	0.892	423	348	82.3	75	17.7	0	0.0	0	0.0	0	0.0
Marmoset/C.j./Primates	31,540	21.06	0.826	26,052	23,890	91.7	1,693	6.5	469	1.8	0	0.0	0	0.0
Mole/S.a./Insectivora	5,900 (14,210)	11.01	0.901	12,803	9,730	76.0	3,073	24.0	0	0.0	0	0.0	0	0.0
Monkey/C.a./Primates	75,310	27.14	0.787	59,269	55,179	93.1	2,252	3.8	1,600	2.7	0	0.0	237	0.4
Monkey/M.f./Primates	46,260	31.01	0.763	35,296	32,331	91.6	1,588	4.5	1,377	3.9	0	0.0	0	0.0
Monkey/M.m./Primates	113,230	34.50	0.743	84,130	82,447	98.0	1,094	1.3	589	0.7	0	0.0	0	0.0
Monkey/S.s./Primates	50,080	23.40	0.810	40,565	36,792	90.7	2,272	5.6	1,420	3.5	0	0.0	81	0.2
Opossum/D.v./Marsupialia	10,095	17.03	0.854	8,621	8,207	95.2	379	4.4	0	0.0	0	0.0	35	0.4
Opossum/M.d./Marsupialia	550	12.31	0.890	490	397	81.1	93	18.9	0	0.0	0	0.0	0	0.0
Rabbit/O.c./Lagomorpha	6,780	22.70	0.815	5,526	4,227	76.5	663	12.0	0	0.0	636	11.5	0	0.0
Raccoon/P.l./Carnivora	127,670	22.22	0.818	104,434	101,928	97.6	2,402	2.3	0	0.0	0	0.0	104	0.1
Slow loris/N.c./Primates	26,360	23.14	0.812	21,404	20,013	93.5	1,049	4.9	342	1.6	0	0.0	0	0.0
Tree shrew/T.g./Scandentia	8,394	20.80	0.828	6,950	6,672	96.0	278	4.0	0	0.0	0	0.0	0	0.0
Vole/P.p./Rodentia	7,120	14.74	0.872	6,209	4,483	72.2	459	7.4	0	0.0	1,267	20.4	0	0.0

¹G.s., Genus specie; see Table 1.²Numbers in parentheses represent the total number of CS somata in ipsilateral hemisphere in species in which the ipsilateral total exceeded the contralateral total. Numbers to the left represent the total number of CS somata in the contralateral hemisphere.

or unlabeled somata were counted that were as large as or larger than the smallest HRP-labeled somata in the sample.

Rostral/caudal ratio within region A. As noted in Nudo and Masterton (1990a), the density of labeled somata is not homogeneous within region A. In each animal, distinct zones of high density are present, commonly in the rostral (i.e., motor) portion of region A. To quantify the degree of differentiation of rostral (motor) and caudal (sensory) zones within region A, the ratio of the density of the rostral peak to the density of the caudal peak was derived.

Condensation index within region A. Because in some species (e.g., opossums) the zones of highest density within region A appeared to be distributed throughout the labeled region as opposed to other species (e.g., Old World monkeys) in which the zones of highest density appeared to be confined to a small part of the region, a "condensation index" was derived. This index is arithmetically defined as the percentage area of region A containing labeled cell densities less than 40% of the peak surface density. Thus, the condensation index is highest if only a small proportion of region A contains a high density of CS somata.

Each of the measurements described, whether direct or derived, along with body weight, brain weight, entire neocortical surface area, and several other descriptive parameters derived from these measurements were then tabulated for each of the 24 species arranged taxonomically. They were then analyzed with conventional parametric statistical procedures to help test a variety of hypotheses regarding their variation due to allometric or scaling factors or their variation due to ancestral history.

RESULTS

The results are divided into five sections. The first describes the quantitative values from direct and derived measurements; the second describes relationships of these values with body size, brain size, and neocortex surface area of the 24 species in the sample; the third describes differences in these relationships between the 7 primate species and the 17 nonprimate species; the fourth describes the

relationships of morphological characteristics to ancestry; and the fifth describes the relationships of morphological characteristics to ancestry with body size, brain size, and neocortical area held constant.

Quantitative results

The six chief quantitative results directly obtained from the histological materials are described first, followed by the seven secondary results derived from these chief results.

Total number of labeled somata and soma diameter. The total number of CS profiles in the neocortex contralateral to the spinal cord hemisection of each of the 24 species and the total number of CS profiles ipsilateral to the spinal cord hemisection in 5 of the 24 species (discussed later) is shown in Table 2. In addition, Table 2 includes the soma diameter and the estimate of the number of CS somata after stereological correction for overcounting (see Nudo and Masterton, 1989). These corrected values are those used in each subsequent analysis.

In 5 of the 24 species, the three hedgehogs, mole, and hyrax, the total number of CS somata ipsilateral to the hemisection exceeded the total number contralateral to the hemisection. In these species, the ipsilateral totals are used in subsequent analyses and illustrations (e.g., Fig. 1). For a further discussion of interhemispheric asymmetry in CS neurons, see Nudo and Masterton (1990a).

After stereological correction, the mean number of CS somata in the 24 species was 24,071. The highest number was found in raccoon (104,434) and the lowest number was found in least shrew (423). When the species are grouped by taxonomic order, the highest average number of CS somata was found in Carnivora (68,736) and the lowest number was found in Insectivora (4,195).

The highest number of CS somata per gram of body weight was found in vole and the lowest number was found in rabbit (Fig. 1). The highest number of CS somata per milligram of brain weight was found in mole and the lowest number was found in gray, short-tailed opossum (Fig. 1).

NUMBER OF CORTICOSPINAL SOMATA, NUMBER PER GRAM BODY WEIGHT, NUMBER PER MILLIGRAM BRAIN WEIGHT AND SOMA DIAMETER RANK ORDERED BY SPECIES

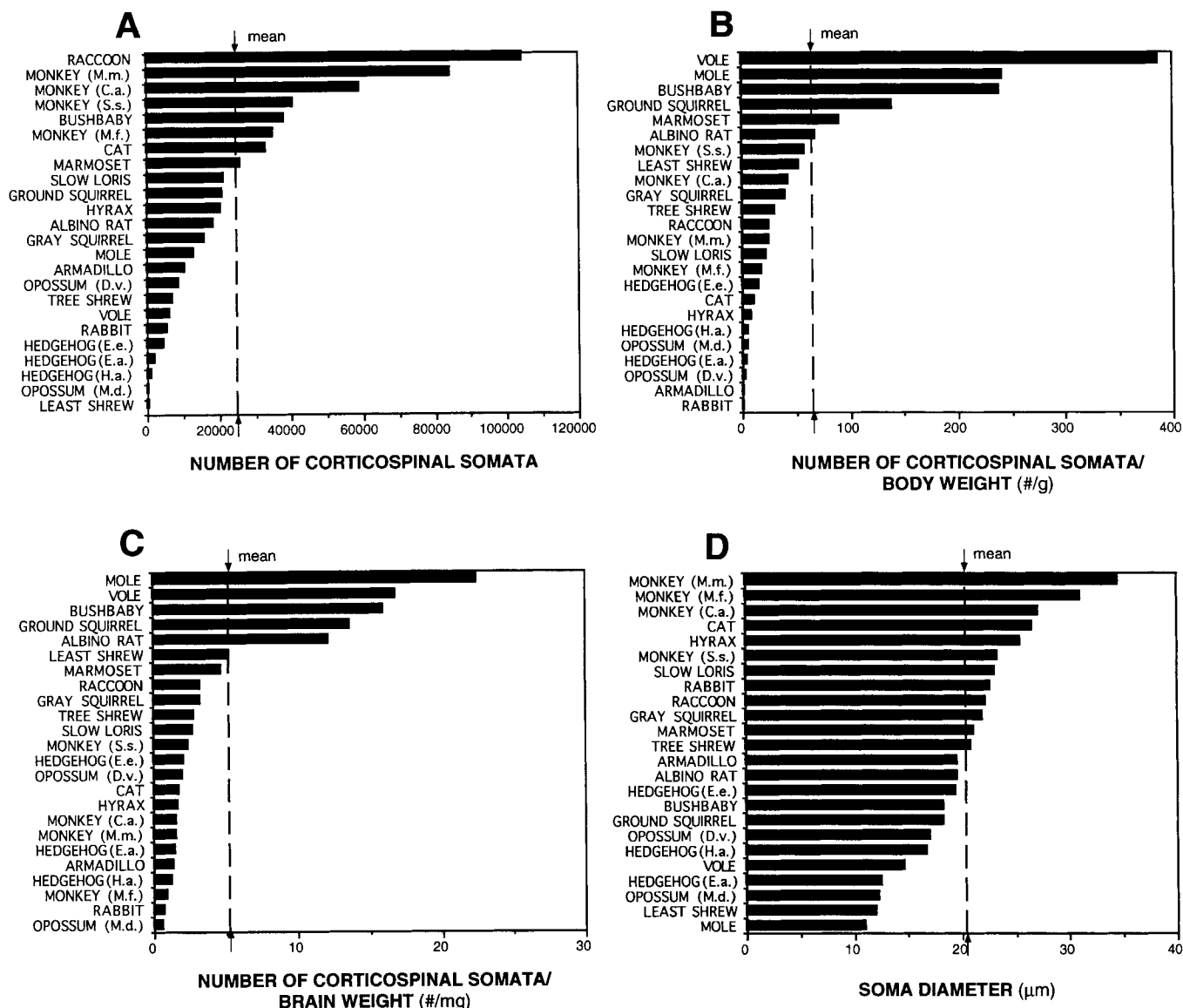


Fig. 1. **A–D:** Graphs show the distribution of number of corticospinal (CS) somata, number of CS somata per gram body weight, number of CS somata per milligram brain weight, and soma diameter in 24 mammals.

The average diameter of the CS somata in the 24 species was 20.48 μm . The largest average soma diameter was found in rhesus macaque (34.5 μm) and the smallest was found in mole (11.01 μm).

Total number and percentage of CS somata in each region. Of the two regions common to each of the 24 species (i.e., regions A and B), the highest number of CS somata was invariably found in region A. Indeed, region A contained several times more CS somata than the other regions combined.

The total number of CS somata in each of the regions for each of the 24 species is listed in Table 2.

Region A. On average, the number of CS somata in region A accounted for 87.6% of the total number of CS somata. The average number of CS somata in region A was 22,235. The highest number was found in raccoon (101,928) and the lowest was found in least shrew (348) paralleling the distribution of the total number of CS somata in these species.

When the species were grouped by taxonomic order, the highest average number of CS somata in region A was found in Carnivora (65,897) and the lowest number was found in Insectivora (3,246). The average number of CS somata, the average number of somata in each region, and

the average percentage number of somata in regions A and B are listed in Table 3 for each of the nine taxonomic orders represented in the sample.

The highest percentage of CS somata in region A was found in macaque monkey (98.0%) and the lowest percentage was found in pine vole (72.2%). When grouped by taxonomic order, the highest percentage of somata in region A was found in Edentata (armadillo, 96.5%) and the lowest percentage was found in Lagomorpha (rabbit, 76.5%).

Region B. The other region common to all species, region B, was the second largest in its total number of CS somata in each species except for most of the rodents. On average, CS somata in region B accounted for 9.0% of the total number of CS somata. The average number of CS somata in region B was 1,241. The highest number was found in the primate, bushbaby (3,294), and the lowest number was found in the insectivore, least shrew (75).

When the species were grouped by taxonomic order, the highest average number of CS somata in region B was found in Carnivora (cat and raccoon, 2,407) and the lowest number was found in Marsupial (opossums, 236).

The highest percentage of CS somata in region B was found in the insectivore, mole (24.0%), and the lowest percentage was found in the primate, macaque monkey (1.3%). When the species were grouped by taxonomic order, the highest percentage of CS somata in region B was found in Insectivora (19.9%) and the lowest percentage was found in Edentata (armadillo, 3.5%).

Regions C and C'. Region C is present in each of the primate species but in no others (Nudo and Masterton, 1990a). In primates, region C accounted for 2.2% of the total number of labeled somata. The average number of CS somata in region C in primates was 888, with the highest number in green monkey (1,600) and the lowest number in slow loris (342).

Region C' is present in each of the rodent species and the rabbit but in no others (Nudo and Masterton, 1990a). Of those species in which it was present, region C' accounted for 11.6% of the total number of CS somata. The average number of CS somata in region C' in these species was 1,319, with the highest number in albino rat (2,119) and the lowest number in rabbit (636).

Maximum surface density of CS somata by region. The maximum surface density of CS somata derived in each of the 24 species is listed in Table 4.

For region A this value averaged 1,367 cells/mm², with the highest in bushbaby (4,030) and the lowest in crab-eating macaque (300) and rhesus macaque (408).

The maximum surface density in region B averaged 421 cells/mm² or about 31% of the maximum density in region A. Maximum density in region B was highest in mole (1,355) and lowest in rhesus macaque (99).

The maximum density in region C for each of the seven primate species was 522 or about 43% of the maximum density in region A and about 33% higher than the maximum density in region B. The maximum density in region C was highest in bushbaby (1,504) and lowest in crab-eating macaque (128).

In the four rodents and the rabbit in which region C' is found, its maximum density averaged 994 cells/mm² or about 49% of the maximum density in region A and about 78% of the maximum density in region B. The maximum density in region C' was highest in rat (1,606) and lowest in rabbit (170).

TABLE 3. Number of Corticospinal Somata by Taxonomic Order

Order	# Species	Mean # CSN ¹	Mean #A	Mean #B	Mean #C	Mean #C'	Mean %A	Mean %B
Carnivora	2	68,736	65,897	2,407	0	0	94.0	4.8
Edentata	1	10,454	10,088	366	0	0	96.5	3.5
Hyracoidea	1	20,638	19,008	1,548	0	0	92.1	7.5
Insectivora	5	4,195	3,246	949	0	0	80.1	19.9
Lagomorpha	1	5,526	4,227	663	0	636	76.5	12.0
Marsupialia	2	4,556	4,302	236	0	0	88.2	11.7
Primates	7	43,575	40,738	1,892	888	0	92.7	5.0
Rodentia	4	15,391	12,962	917	0	1,490	82.0	6.3
Scandentia	1	6,950	6,672	278	0	0	96.0	4.0

¹Numbers of CS somata are stereologically corrected values.

TABLE 4. Maximum Surface Density of Corticospinal Somata (cells/mm²)

Animal/G.s./Order	Region			
	A	B	C	C'
Albino rat/R.n./Rodentia	2,250	592	0	1,606
Armadillo/D.n./Edentata	736	201	0	0
Bushbaby/G.s./Primates	4,030	1,323	1,504	0
Cat/F.c./Carnivora	1,677	189	0	0
Gray squirrel/S.c./Rodentia	1,664	574	0	976
Ground squirrel/C.t./Rodentia	3,310	602	0	1,143
Hedgehog/E.a./Insectivora	848	326	0	0
Hedgehog/E.e./Insectivora	771	356	0	0
Hedgehog/H.a./Insectivora	676	184	0	0
Hyrax/P.c./Hyracoidea	1,148	446	0	0
Least shrew/C.p./Insectivora	706	107	0	0
Marmoset/C.j./Primates	1,305	435	638	0
Mole/S.a./Insectivora	3,472	1,335	0	0
Monkey/C.a./Primates	900	228	464	0
Monkey/M.f./Primates	300	128	128	0
Monkey/M.m./Primates	408	99	161	0
Monkey/S.s./Primates	760	253	535	0
Opossum/D.v./Marsupialia	857	429	0	0
Opossum/M.d./Marsupialia	655	393	0	0
Rabbit/O.c./Lagomorpha	540	256	0	170
Raccoon/P.l./Carnivora	786	171	0	0
Slow loris/N.c./Primates	875	282	226	0
Tree shrew/T.g./Scandentia	1,790	429	0	0
Vole/P.p./Rodentia	2,339	759	0	1,075

Average surface density of CS somata. The average surface density for each of the species is listed in Table 5. The mean average density for the 24 species was 260 cells/mm² with the highest in mole (587) and bushbaby (583) and the lowest in crab-eating macaque (62), which is more than a ninefold difference.

Maximum thickness of layer of CS somata. Although all CS somata were contained within layer V in every case, only a small percentage of somata within layer V were labeled (see Concentration of CS Somata). Therefore, the thickness of the layer of CS somata was rarely equivalent to the thickness of layer V.

The layer of CS somata within layer V was invariably thickest in the zone of maximum density within region A. Large variations were found in the maximum thickness in the 24 species (see tabulated values in Table 5). The highest value was found in slow loris (0.71 mm) and the lowest was found in short-tailed opossum (0.10 mm), a sevenfold difference.

Maximum volume density of CS somata. Because of the large differences in the thickness of the CS layer among species, a measure of the density of CS somata per unit of volume of layer V was derived (Table 5). The average maximum volume density was 5,606 cells/mm³, with the highest in mole (17,848) and lowest in rhesus macaque (1,196), nearly a 15-fold difference.

Maximum column height of CS somata. Because CS somata often appeared to be vertically arranged, one above another in the same focal plane within layer V, and because the maximum number of CS somata in these columns was

TABLE 5. Nine Other Morphological Characteristics of Corticospinal Somata

Animal/G.s./Order	Avg surface density (cells/mm ²)	Thickness (μm)	Maximum volume density (cells/mm ³)	Column height (cells)	Concentration (%)	Concentration (large cells; %)	Rostral/caudal	Condensation	% Medial-lateral distribution
Albino rat/R.n./Rodentia	639	0.38	6,475	5.02	26.7	78.7	0.89	93.0	29
Armadillo/D.n./Edentata	140	0.18	4,089	2.51	35.8	n/a	0.80	84.7	29
Bushbaby/G.s./Primates	830	0.38	11,918	7.61	34.4	91.6	3.23	96.9	27
Cat/F.c./Carnivora	173	0.17	11,526	2.37	36.7	69.0	6.67	99.4	65
Gray squirrel/S.c./Rodentia	414	0.36	5,283	3.28	28.3	73.5	2.86	95.0	30
Ground squirrel/C.t./Rodentia	800	0.38	9,751	6.76	36.8	89.8	1.10	93.7	33
Hedgehog/E.a./Insectivora	257	0.17	5,432	2.66	18.8	42.9	0.85	94.1	56
Hedgehog/E.e./Insectivora	195	0.35	2,640	5.02	28.4	55.2	0.93	92.7	32
Hedgehog/H.a./Insectivora	138	0.19	3,965	2.57	15.7	61.0	1.22	96.8	38
Hyrax/P.c./Hyracoidea	258	0.35	3,280	4.00	n/a	n/a	1.39	84.5	64
Least shrew/C.p./Insectivora	245	0.15	5,116	2.68	10.7	n/a	1.00	93.1	50
Marmoset/C.j./Primates	355	0.35	4,248	4.13	21.8	80.5	1.82	98.5	33
Mole/S.a./Insectivora	802	0.21	17,848	7.21	34.1	52.6	1.41	92.2	35
Monkey/C.a./Primates	106	0.37	2,849	2.36	19.6	63.3	5.26	99.0	78
Monkey/M.f./Primates	76	0.26	1,372	2.29	9.1	79.5	10.00	98.9	54
Monkey/M.m./Primates	108	0.41	1,196	2.23	13.2	86.1	1.64	94.3	78
Monkey/S.s./Primates	178	0.38	2,302	2.43	13.9	69.7	2.70	94.6	68
Opossum/D.v./Marsupialia	298	0.20	4,781	2.56	23.1	65.2	1.16	83.7	55
Opossum/M.d./Marsupialia	168	0.10	7,125	2.67	15.9	43.4	1.25	78.7	54
Rabbit/O.c./Lagomorpha	116	0.46	1,948	2.45	20.8	71.4	2.13	94.8	39
Raccoon/P.l./Carnivora	103	0.38	2,368	3.27	27.9	87.5	3.23	98.1	61
Slow loris/N.c./Primates	263	0.71	1,417	4.06	12.6	79.1	3.45	94.0	56
Tree shrew/T.g./Scandentia	474	0.31	6,555	4.97	28.7	76.9	2.86	90.9	64
Vole/P.p./Rodentia	685	0.33	7,821	6.10	26.3	81.8	1.18	91.0	40

variable among the animals, a measure of column height of CS somata was derived. The maximum column height observed in each of the species is also listed in Table 5. The average column height for the 24 species was 3.85 cells (stereologically corrected). Column height was highest in bushbaby (7.61) and lowest in macaque monkey (2.23), more than a threefold difference.

Concentration of CS somata. The concentration of the CS somata, a derived measure of the number of CS somata divided by the total number of somata subjacent to a 1 mm × 50 μm sample of cortex, also varied among the animals and is tabulated in Table 5. The mean concentration was 22.9%, with the highest in ground squirrel (36.8%) and the lowest in crab-eating macaque (9.1%), a fourfold difference.

The CS somata were invariably the largest somata in layer V of each region. Therefore, unlabeled (non-CS) somata approaching the size of the largest labeled (CS) somata were only rarely observed. When only those somata that were as large or larger than the smallest HRP-labeled somata in the sample were counted and the number used as denominator, the concentration was much higher (see Table 5). Using this method of measuring the concentration of largest layer V somata, the average concentration was 71.4%, with the highest value in bushbaby (91.6%) and the lowest value in gray, short-tailed opossum (43.4%).

Rostral/caudal ratio within region A. The density of CS somata within a given region was not homogeneous. Within region A, for example, there was considerable variation among the species in the location of relatively high-density and low-density zones. In most cases, a single, high-density zone (> 1 S.D. above the mean density) occupied more neocortical surface area than other high-density zones within region A. This high-density zone was usually located in the rostral half of region A. To quantify this feature, the peak surface density in the rostral half and the caudal half of region A was measured first. Then the rostral/caudal ratio was derived (Table 5).

The highest rostral peak/caudal peak ratio was found in crab-eating macaque (10.00), and the lowest ratio was found in West African hedgehog (0.85), nearly a 12-fold difference. In primates, the high-density zone in the rostral

half of region A was in the anterior bank of the central sulcus, extending onto its exposed superior lip (i.e., area 4), and the more rostral and lateral lower density zone was probably in area 6.

Condensation index within region A. Because in some species (e.g., opossums) the zones of highest density within region A appeared to be distributed throughout the labeled region as opposed to other species (e.g., Old World monkeys) in which the zones of highest density appeared to be confined to a small part of the labeled region, a condensation index was derived to quantify this characteristic (see Table 5). The condensation was highest in the green monkey (99.0%) and crab-eating macaque (98.9%) and lowest in gray, short-tailed opossum (78.7%).

Medial-lateral position of zone of maximum density within region A. The medial-lateral position of the zone of highest density within region A was measured in each species and is listed in Table 5 as a percentage of the entire medial-lateral dimension of region A. The zone of highest density was most medial in bushbaby (27%) and most lateral in rhesus macaque and green monkey (78%).

Allometric, or scaling, relationships

Because the animals employed in this study varied over a wide range of body size (9 g for least shrew to over 5 kg for rabbit), it was possible to identify those CS soma characteristics that varied reliably with the size of the animal. Body weight, brain weight, and total neocortical surface area for each of the 24 species are listed in Table 1.

These three size variables are not independent. For example, log body weight and log brain weight are closely related (e.g., Jerison, 1973). Here, we can add that log brain weight and log neocortical surface area are also very closely related (see Fig. 2). Log brain weight accounts for 95% of the variance in log neocortical surface area. As a result, one would expect that the relationships of most of the quantitative characteristics with brain weight are closely similar to their relationships with neocortical surface area (e.g., Hofman, 1982, 1985). However, because the differences between relationships with brain weight and relationships

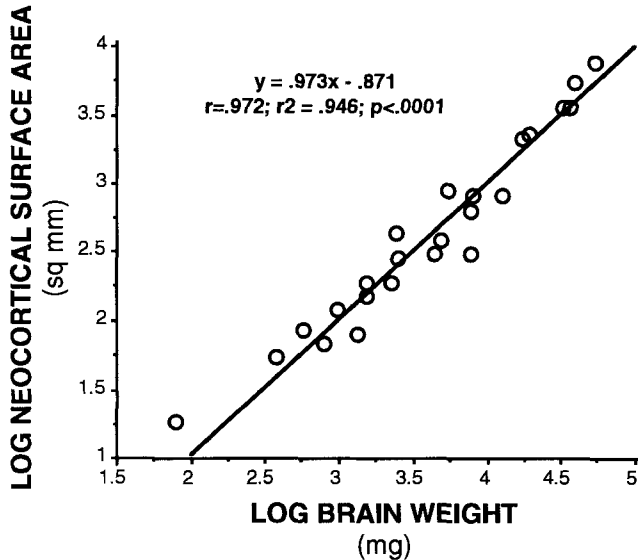


Fig. 2. Graph shows relation of neocortical surface area to brain weight in 24 species of mammals. Although log brain weight accounts for 95% of the variance in log neocortical surface area ($r^2 = 0.946$), several important differences were found in their relationships with CS characteristics. The slope in this and each of the following graphs is based on "reduced major-axis" analysis.

with neocortical surface area that do occur may reflect important differences between cortical and noncortical structures, relationships both with brain weight and with neocortical surface area are included in the statistical analyses.

Using conventional multivariate procedures for examination of allometric or scaling factors, an equation for simple allometry was derived for each of the 23 soma characteristics versus body weight, brain weight, and neocortical surface area. As is the convention, the values for each of the 23 characteristics were first transformed to their logarithms (e.g., see Gould, 1966) so that the allometry equation became linear with the general form

$$\log y = b_1^* X_1 + b_2^* X_2 + b_3^* X_3 + b_0$$

where y is one of the 23 CS characteristics, X_i is one of the size variables, b_i is the partial regression coefficient for size parameter i , and b_0 is the Y intercept. Regression coefficients calculated with the logarithmic transformation were not substantially different from coefficients obtained with other proposed transformations, such as square root transformations for counts, and arcsin transformations for percentages and proportions (Bartlett, 1936; Zar, 1984). Slopes of bivariate relationships are derived from reduced major axis equations (e.g., Seim and Saether, 1983; Hofman, et al., 1986).

Results of multiple regression analyses derived for each of the 23 soma characteristics are listed in Table 6. These results show that 11 of the characteristics were reliably related to size. Table 6 also lists standard coefficients, which provide a rough estimate of the weighting for each of the three variables. However, because of the covariation among body weight, brain weight, and neocortical surface area, a second multivariate approach, a "stepwise regression," was then used to determine which, if any, of the three size

variables were most useful in predicting each of the soma characteristics.

The stepwise regression procedure began with each of the three size variables in the regression model. Then, the partial F ratio for each variable was examined. If the smallest F ratio was less than a preset criterion ($F_{\text{to-remove}} = 4.0$), the corresponding variable was removed from the model. This procedure continued until the smallest F ratio was greater than the criterion ("backward" procedure). The final model included the minimum number of size variables that could still account for variation in the dependent variable. For example, when all three size variables were included in the model, 71% of the variation in the total number of CS somata could be accounted for by size. Stepwise regression demonstrated that a single size variable, neocortical surface area, accounts for 69% of the variation in the number of CS somata.

The models derived from stepwise regression revealed that a total of 18 characteristics were related to one or more size variables. In fact, excluding those CS characteristics found in only a subset of the entire sample (i.e., characteristics related to regions C and C'), only one characteristic, concentration, was unrelated to size.

Relation of CS somata to body weight. Based on models derived from stepwise regression, body weight was the best predictor of four soma characteristics: density in regions A, B, and C, average surface density, and column height (Table 6). Each of these relationships was negative, i.e., mammals with increased body size tended to have CS somata that were less densely distributed when projected to the surface. For example, body weight alone accounted for 54% of the variation in average surface density ($r = 0.734$, $P < 0.0001$). Bivariate plots of selected relationships are shown in Figures 3 and 4.

Relation of CS somata to brain weight. Again, based on stepwise regression, brain weight was the best predictor of two soma characteristics: number of CS neurons in region C and medial-lateral position. Each of these relationships was positive (Table 6), i.e., with increased brain size, mammals in the sample tended to have more CS neurons in region C and CS neurons that were more lateral. For example, brain weight alone accounted for 21% of the variation in medial-lateral position ($r = 0.461$, $P = 0.0233$). Brain weight, with a second predictor, body weight, accounted for 81% of the variation in the number of neurons in region C ($r = 0.900$, $P = 0.0362$), i.e., primates with large brains but small bodies tend to have more neurons in region C.

Relation of CS somata to neocortical surface area. Neocortical surface area was the best predictor of 11 soma characteristics (Table 6). Stepwise regression identified surface area as the sole reliable predictor of eight characteristics. Positive relationships were found for surface area and the following six characteristics: number of CS somata, number in regions A and B, percentage in region A, soma diameter, and layer thickness. For example, surface area alone accounted for 69% of the variation in the number of CS somata ($r = 0.832$, $P < 0.0001$) and 82% of the variation in their soma diameter ($r = 0.907$, $P < 0.0001$). Negative relationships were found for surface area and the following two characteristics: percentage in region B and volume density.

Furthermore, stepwise regression identified surface area as best predictor of three other characteristics, i.e., surface area combined with brain weight were reliable predictors of

TABLE 6. Relation of Corticospinal (CS) Somata Characteristics to Body Weight, Brain Weight, and Neocortical Surface Area in 24 Mammals

Log characteristic	Multiple regression					Stepwise regression					
	R ²	Standard coefficient			F ratio	R ²	Standard coefficient			Slope ¹	
		Body weight	Brain weight	Surface area			Body weight	Brain weight	Surface area		
# CS somata	0.710	-0.293	0.289	0.778	16.285**	0.692			0.832	49.454**	0.928
#A	0.721	-0.271	0.287	0.771	17.246**	0.707			0.841	53.108**	0.961
#B	0.493	-0.351	0.056	0.888	6.495**	0.449			0.670	17.941**	0.668
#C	0.853 ²	-1.953	3.687	-1.200	5.815	0.810	-1.787	2.350		8.518*	0.401
#C'	0.765	-0.268	7.513	-7.719	1.088	—				—	—
%A	0.442	0.210	0.151	0.335	5.276**	0.415			0.644	15.594**	0.056
%B	0.479	0.079	-0.489	-0.274	6.133**	0.473			-0.688	19.782**	-0.471
%C	0.385	-1.609	3.875	-2.177	0.627	—				—	—
%C'	0.482	1.262	4.098	-5.720	0.311	—				—	—
density A	0.205	-0.514	0.226	-0.176	1.721	0.204	-0.452			5.651*	-0.379
density B	0.227	-0.749	1.653	-1.433	1.955	0.165	-0.407			4.360*	-0.391
density C	0.888	-1.703	1.641	-0.812	7.936	0.807	-0.898			20.875**	-0.485
density C'	0.702	-0.987	3.651	-3.378	0.784	—				—	—
surface density	0.591	-0.414	-0.116	-0.280	9.648**	0.538	-0.734			25.634**	-0.426
soma diameter	0.826	-0.008	0.248	0.673	31.651**	0.822			0.907	101.918**	0.186
layer thickness	0.238	-0.235	0.272	0.396	2.085	0.228			0.478	6.513*	0.273
volume density	0.371	-0.328	0.082	-0.403	3.931*	0.334			-0.578	11.049**	-0.464
column height	0.214	-0.542	0.563	-0.523	1.815	0.204	-0.452			5.645*	-0.225
concentration	0.043	0.057	0.599	-0.789	0.269	—				—	—
concentration (lg somata)	0.460	0.347	-3.427	3.571	4.832*	0.441		-2.520	2.951	7.115**	0.139
condensation	0.455	0.200	-2.671	2.873	5.563**	0.451		-2.210	2.579	8.609**	0.038
med-lat position	0.238	-0.223	0.389	0.276	2.080	0.213		0.461		5.943*	0.210
rost/caud density ratio	0.613	-0.222	-0.247	-0.701	10.557**	0.606		-1.271	1.956	16.168**	0.434

¹Slope of "best" predictor for reliable relationships.²Values for regions C and C' are italicized since these regions exist in a small subset of the species.* $P < 0.05$; ** $P < 0.01$.

concentration of large somata, condensation, and rostral/caudal ratio. In each of these three relationships, surface area was positively related, whereas brain weight was negatively related to the characteristics.

Differences in allometric relationships between primates and other mammals

Several of the relationships between the 23 CS characteristics and the sizes of body, brain, and neocortex were similar both in primates and in nonprimates (Table 7). For example, soma diameter was positively and reliably related to neocortical surface area (the best predictor based on multiple or stepwise regression analysis) in primates ($r = 0.964$, $P = 0.0005$) and in nonprimates ($r = 0.847$, $P < 0.0001$). Average density was negatively and reliably related to body weight in primates ($r = -0.949$, $P = 0.0011$) and in nonprimates ($r = -0.715$, $R^2 = 0.511$, $P = 0.0013$).

Although similar relationships were observed for several CS characteristics, other relationships differed between primates and nonprimates. For example, one characteristic, medial-lateral position, was best predicted by brain weight. However, this relationship was valid only for primates ($r = 0.883$, $P = .0085$) but *not* for nonprimates ($r = 0.229$, $P = 0.3776$). Three characteristics, density in region A, density in region B, and column height, were negatively, but reliably, related to body weight in primates ($r = -0.907$, $P = 0.0049$; $r = -0.959$, $P = 0.0006$; $r = -0.864$, $P = 0.0122$, respectively) but were unrelated to body weight in nonprimates ($r = -0.340$, $P = 0.1811$, $r = -0.268$, $P = 0.2977$, $r = -0.361$, $P = 0.1550$, respectively). Conversely, other characteristics (number of CS somata, number in region A, and number in region B) were reliably related to neocortical surface area in nonprimates but *not* in primates.

Furthermore, some characteristics were reliably related to the size of body, brain, or neocortex both in primates and in nonprimates, but the slopes of their regression lines were significantly different. Because differences in these slopes may reflect structural and/or functional divergence due to

differential increases or decreases in the particular CS characteristic, each was considered in more detail.

First, surface density was negatively, but reliably, related to body weight (best predictor) in both primates and nonprimates (Table 7, Fig. 4). However, surface density decreased by a greater amount in primates with increased body weight, i.e., the slope of the regression line for primates (-0.766) was two times greater than that for nonprimates (-0.348), and this difference was statistically reliable ($t = 8.67$, $P < 0.001$).

Second, concentration was positively related to surface area (best predictor) in nonprimates but was unrelated to surface area (or to body weight or brain weight) in primates (Table 7, Fig. 4). Nevertheless, it is likely that the slope of the regression line is negative in primates, as shown in Figure 4F.

Third, soma diameter was positively related to neocortex size (best predictor) both in primates and in nonprimates (Table 7, Fig. 3). The slope of the regression line relating soma diameter to neocortical size was slightly, but reliably, greater in primates than in nonprimates (slopes = 0.206 and 0.200, respectively; $t = 3.0$, $P < 0.01$; Table 7), but this difference is probably too small to be biologically meaningful.

Finally, it is likely that for other scaling relationships, the elevations of the regression lines differ. For example, Figure 3 shows that the numbers of CS somata in each of the seven primates lie on or above the mammalian regression line relating number and body weight. In contrast, primate values lie on both sides of the regression line relating number and neocortical surface area. Thus, primates probably possess more CS somata than body weight suggests but not more than cortical area suggests. Similarly, Figure 3 shows that the diameters of CS somata in the seven primates lie above the mammalian regression line relating soma size and body weight. In contrast, the primate values lie on or below the nonprimate regression line relating soma size and neocortical surface area. Thus,

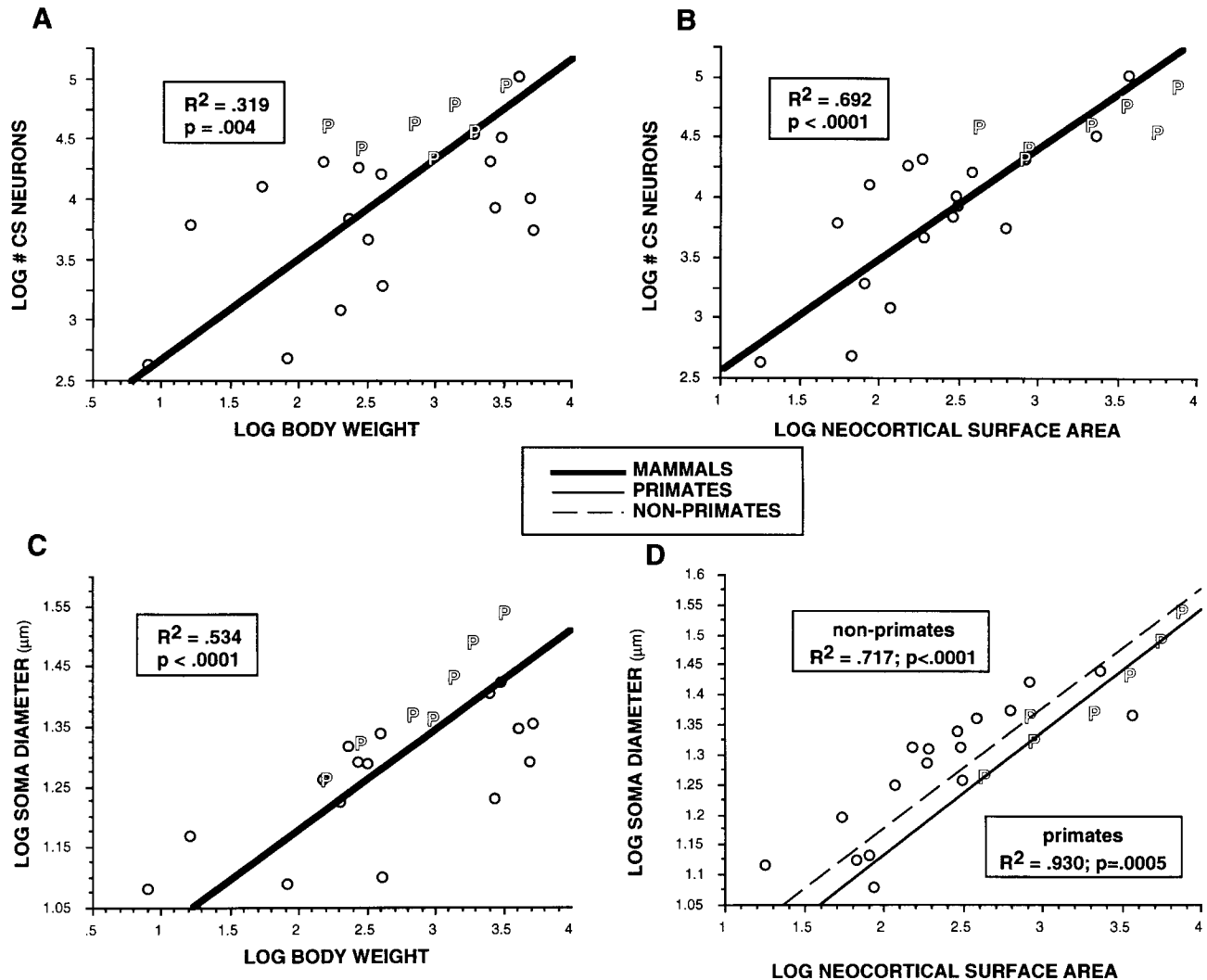


Fig. 3. A-D: Graphs show relationship between total number of CS somata and soma diameter with body size and neocortex size in 24 species of mammals. Stereologically corrected numbers of CS somata are logarithmically transformed for statistical purposes. Thick solid line indicates reduced major-axis regression lines for all 24 mammals; thin solid line indicates statistically reliable linear regression lines for seven primates; dashed line indicates statistically reliable linear regres-

sion lines for 17 nonprimates. Although the primate regression coefficients for number versus body weight and soma diameter versus body weight were not reliably different from those of nonprimates, each of the values for the seven primates lie on or above the mammalian regression line. This indicates that the number of CS somata is higher and the diameter of CS somata is larger in primates for any given body weight. P, primate species.

primates probably possess larger CS somata than body weight suggests but not more than cortical area suggests.

Relation of morphological characteristics of CS somata to the ancestry of Old World monkey, raccoon, and gray squirrel

We now turn to the relationship between the animal's ancestral history and each of the quantitative morphological and cytological characteristics. The derivation of kinship orderings from cladistic relationships has been described elsewhere (e.g., Nudo and Masterton, 1989). Briefly, we have chosen three of the 24 possible kinship orderings represented in our sample of mammals. Besides Old World monkey, we have chosen orderings of kinship with raccoon because of similarity in its manual dexterity with primates and because it has the largest number of CS somata among

the animals in the present sample, and gray (tree) squirrel because of its similarities with the visual system of primates.

Although the previous section showed that a large portion of the variation in CS characteristics can be accounted for by scaling factors, we first describe the relation of the values with kinship uncorrected for size. This allows the identification of differences between the Old World monkey lineage and the raccoon or gray squirrel lineage. Then, in the following section, each of the scaling factors is held mathematically constant to observe their effects on these differences.

Kinship with Old World monkey. Of the 23 morphological characteristics that were examined for relationship with the Old World monkey lineage, seven showed statistically reliable increases and four showed reliable decreases with

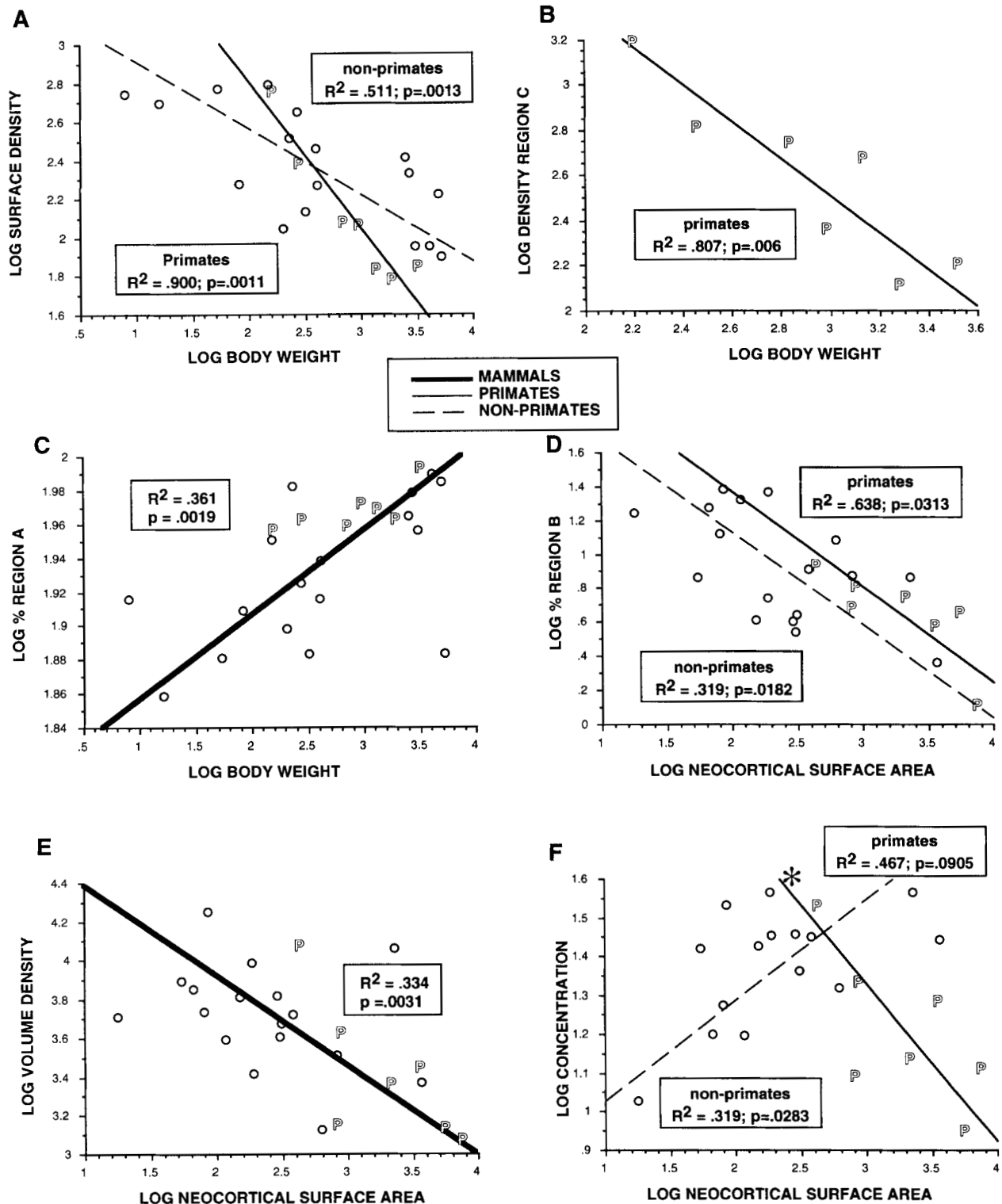


Fig. 4. A–F: Graphs show relationship between selected CS characteristics and their “best” predictors in 24 species of mammals. Symbols are the same as in Figure 3. Asterisk in F indicates that correlation is not statistically reliable. Regression line is included to show probable trend.

increasing kinship with Old World monkey. The seven characteristics that increased were: total number of CS somata, the number in regions A and B, soma diameter,

thickness of CS somata layer within layer V of the cortex, condensation, and rostral/caudal ratio. The four characteristics that decreased were: percentage in region B, surface

TABLE 7. Relation of Corticospinal (CS) Somata Characteristics to Best Predictor: Nonprimates Versus Primates

Log characteristic	Best predictor	Nonprimates (n = 17)		Primates (n = 7)	
		R ²	Slope	R ²	Slope
#CS somata	surface area	0.621**	1.087	0.469	—
#A	surface area	0.638**	1.122	0.484	—
#B	surface area	0.449**	0.817	0.154	—
%A	surface area	0.300*	0.072	0.353	—
%B	surface area	0.319*	-0.548	0.638*	-0.563
Density A	body weight	0.116	—	0.823**	-0.792
Density B	body weight	0.072	—	0.920**	-0.803
Surface density	body weight	0.511**	0.348	0.900**	-0.766
Soma diameter	surface area	0.717**	0.200	0.930**	0.206
Layer thickness	surface area	0.181	—	0.161	—
Volume density	surface area	0.110	—	0.536	—
Column height	body weight	0.130	—	0.746*	-0.436
Concentration	surface area	0.319*	0.259	0.466	—
Concentration (large somata)	surface area	0.228	—	0.108	—
Condensation	surface area	0.065	—	0.003	—
Medial-lateral position	brain weight	0.052	—	0.780**	0.359
Rostral/caudal density ratio	surface area	0.490**	0.435	0.056	—

*P < 0.05; **P < 0.01.

TABLE 8. Relation of Corticospinal (CS) Somata in Three Ancestral Lineages

Characteristic	r Values for kinship with		
	Old World monkey	Raccoon	Gray squirrel
#CS somata	0.586**	0.476*	0.163
#A	0.588**	0.472*	0.140
#B	0.482*	0.459*	0.199
#C	0.705	—	—
#C'	—	—	0.169
%A	0.389	0.215	-0.284
%B	-0.470*	-0.286	-0.038
%C	0.057	—	—
%C'	—	—	-0.556
Density A	-0.320	0.062	0.382
Density B	-0.359	-0.262	0.263
Density C	-0.555	—	—
Density C'	—	—	0.238
Surface density	-0.456*	-0.247	0.290
Soma diameter	0.647**	0.309	0.111
Layer thickness	0.419*	0.262	0.410*
Volume density	-0.514*	-0.086	0.109
Column height	-0.271	0.008	0.275
Concentration	-0.458*	0.261	0.340
Concentration (large somata)	0.355	0.379	0.378
Condensation	0.562**	0.595**	0.308
Medial-lateral position	0.331	0.132	-0.459*
Rostral/caudal density ratio	0.549**	0.369	-0.004

*P < 0.05; **P < 0.01.

density, volume density, and concentration (Table 8). Several of these relationships are illustrated graphically in Figures 5 and 6.

Kinship with raccoon. Four of the 23 characteristics showed statistically reliable increases with increasing kinship with raccoon: total number of CS somata, number in regions A and B, and their condensation within region A (Table 8, Figs. 5–6).

Kinship with gray squirrel. Only one of the 23 characteristics, thickness of the CS somata layer within layer V of cortex, showed a statistically reliable increase with increasing kinship with gray squirrel. A second characteristic, medial-lateral position, showed a statistically reliable decrease, i.e., the zone of highest density became progressively more medial (Table 8).

Evolutionary changes in CS somata. Given the reliable changes in some CS characteristics in each of the three ancestral lineages tested, several evolutionary changes in the characteristics of CS somata can be suggested. Figures 7–9 show the suggested morphological changes along the three lineages. These graphs differ somewhat from the

relative kinship (or propinquity) graphs in Figures 5–6 because they show the number of CS somata averaged over the entire set of progeny of each hypothetical ancestor (or branch point) along each lineage (Nudo and Masterton, 1989). In a statistical sense, these averages are the unbiased estimates of the characteristic for each hypothetical common ancestor. Therefore, the graphs in Figures 7–9 indicate the minimum number or most parsimonious morphological changes along each of the three lineages.

These analyses show that changes in the Old World monkey ancestral lineage probably should be described in two stages: (1) a preprosimian stage (kinship levels 1–3) and (2) a postprosimian stage (kinship levels 3–5). The two-stage description is necessary because for some CS characteristics the direction of change differed for the two stages. For other characteristics the apparent rate of change differed substantially between the two stages. For example, the total number of CS somata increased by 81% in the preprosimian stage (due to an 83% increase in region A, a 243% increase in region C, and a 52% increase in region B), whereas the total number of CS somata increased by a far smaller amount, 37%, in the postprosimian stage (due to a 39% increase in region A, a 33% increase in region C, and a 13% decrease in region B; Fig. 7A,C). In terms of the percentage of total CS somata, it appears that regions A and C gradually increased in hypothetical ancestors of Old World monkey (Fig. 7B) at the expense of region B, which gradually decreased (Fig. 7D).

In comparing the three lineages, it is noteworthy that the increase in CS somata in the Old World monkey lineage was not unique. The total number of CS somata also increased in the raccoon lineage and at a substantially faster rate versus the gray squirrel lineage in which the number decreased slightly.

However, the decline in the absolute number of CS somata in region B may be unique to the postprosimian stage of the Old World monkey lineage. In contrast, the number of CS somata in region B appears to remain constant or to increase in the raccoon and gray squirrel lineages (Fig. 7C).

The soma diameter of CS neurons seems to have increased substantially in the Old World monkey lineage—by 24% in the preprosimian stage and by 21% more in the postprosimian stage (Fig. 8A). The soma diameter remained relatively constant in the raccoon and gray squirrel lineages. In contrast, the concentration of the CS somata decreased throughout the Old World monkey lineage but increased slightly in the raccoon and gray squirrel lineages (Fig. 8B).

It is noteworthy that average surface density and volume density of CS somata (i.e., CS somata per cortical surface area or per unit volume of cortex) both declined substantially throughout the Old World monkey lineage (Fig. 8C,D). Average surface area density decreased by 28% in the preprosimian stage and by 63% more in the postprosimian stage, whereas volume density decreased by 34% in the preprosimian stage and by 50% more in the postprosimian stage. Decreases in average surface density and volume density also were seen in the raccoon lineage. That this decrease is not a necessary consequence of the nature of the ratios can be seen in the gray squirrel lineage, where average density and volume density first increased and then decreased.

The thickness of the layer of CS somata first increased by 33% in the preprosimian stage, then decreased by 25% in

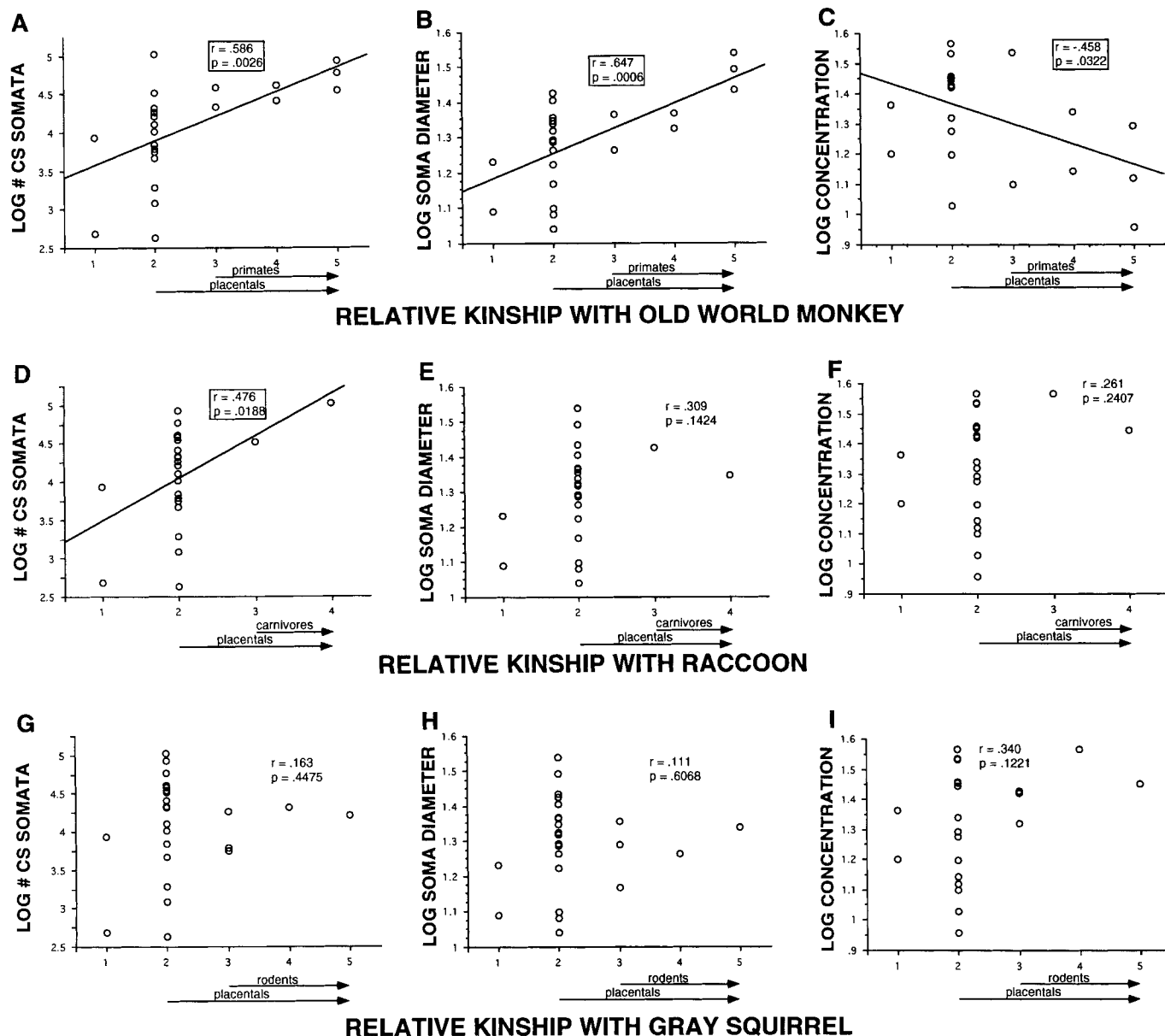


Fig. 5. A–I: Graphs show relationship between morphological characteristics of CS somata (total number, soma diameter, and concentration) and recency of common ancestry with Old World monkey, raccoon, and gray squirrel. Total number of CS somata increased significantly in both the Old World monkey and the raccoon lineage. Soma diameter increased significantly only in the Old World

monkey lineage. Concentration decreased significantly only in the Old World monkey lineage. Statistically reliable relationships are indicated by linear regression lines and boxes enclosing correlation coefficients and *P* values. Because kinship values are fixed, regression lines are based on ordinary linear regression. Slopes are indicative of general trends, not actual rates of change.

the postprosimian stage (Fig. 9A). A related characteristic, the height of columnar stacking of the CS somata, decreased by 7% in the preprosimian stage and then decreased by 36% more in the postprosimian stage (Fig. 9B). In the raccoon lineage, in contrast, the thickness of the labeled layer decreased then increased (Fig. 9A), whereas a modest overall decrease was observed for column height (Fig. 9B). In the gray squirrel lineage, thickness and column height first increased, then decreased.

Finally, condensation, rostral/caudal peak density and medial-lateral position increased throughout the Old World monkey lineage (Fig. 9C,D), i.e., the peak density of CS somata moved rostrally and laterally. A modest shift in medial to lateral position was observed in the raccoon lineage, whereas a shift in the opposite direction, from

lateral to medial, was seen in the gray squirrel lineage. In the raccoon lineage, rostral/caudal peak density first increased (i.e., shifted in density rostrally toward motor cortex), then decreased (i.e., shifted back toward sensory cortex). In the gray squirrel lineage, rostral/caudal peak density first decreased (shifted caudally), then increased (shifted rostrally).

Relation of CS morphology to ancestry with body size, brain size, and neocortical area held constant

Although the relationships between many of the quantitative characteristics of CS somata and the ancestry of the

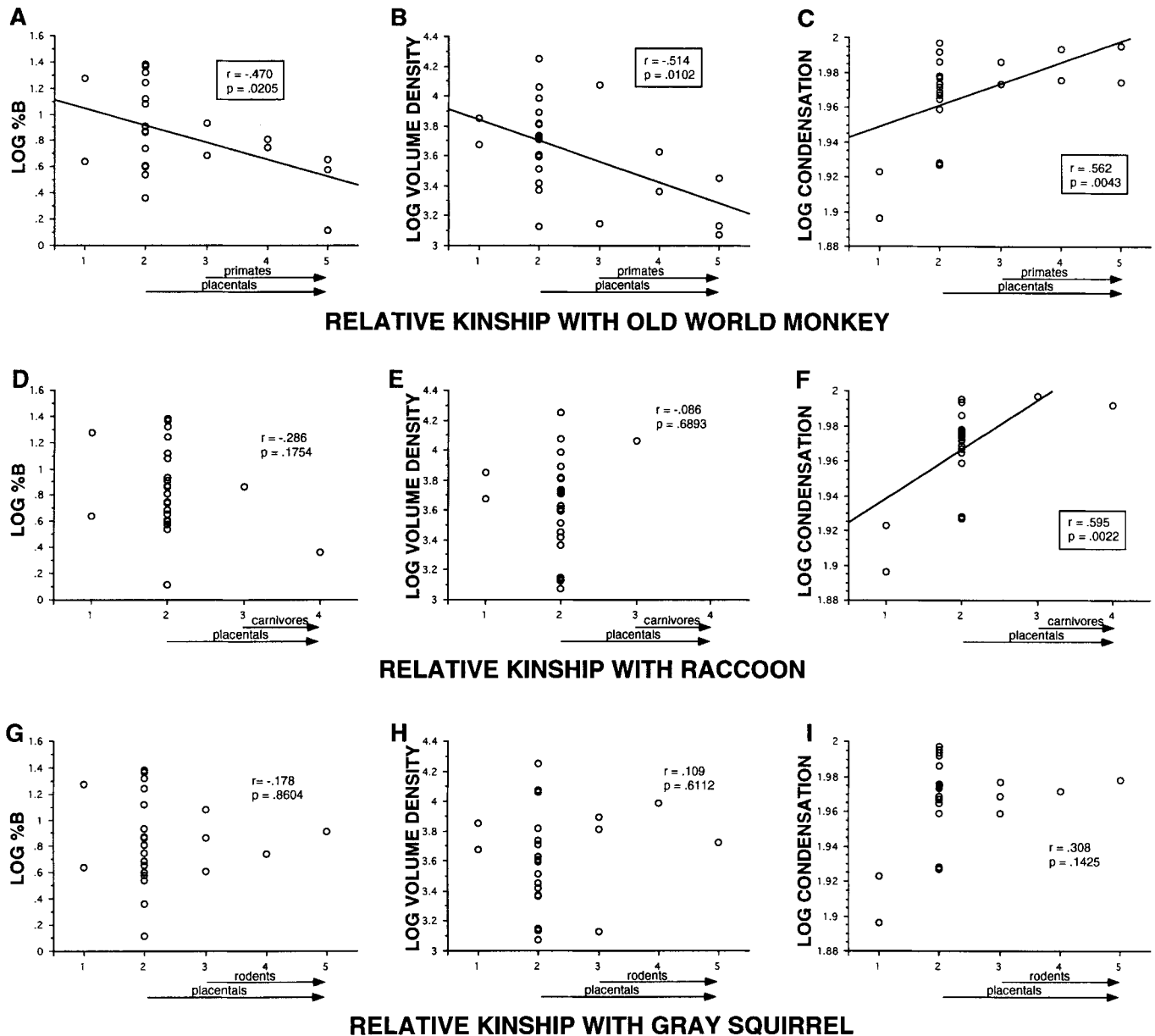


Fig. 6. **A–I:** Graphs show relationship between morphological characteristics of CS somata (percentage of CS somata in region B, volume density, and condensation) and recency of common ancestry. Percentage of CS somata in region B and volume density decreased

significantly only in the Old World monkey lineage. Condensation increased significantly in both the Old World monkey and the raccoon lineages. Statistically reliable relationships are indicated by regression lines and boxes enclosing correlation coefficients and P values.

host are high and reliable, they are complicated by the close relation of each of the same characteristics to body, brain, and neocortical size. However, by using "partial correlation" procedures (e.g., Dunn and Clark, 1974) allowed by the larger sample size available in the present experiments, the relations of these characteristics to ancestry can be examined after the effects of their covariation with body weight, brain weight, and neocortical surface area are held mathematically constant.

Relations of morphological characteristics of CS somata to the Old World monkey ancestral lineage that are independent of body, brain, and neocortical size. When body size was held statistically constant, six of the 23 characteristics still showed a reliable increase with increas-

ing kinship with Old World monkey (Table 9): number of CS somata, number in regions A and B, soma diameter, condensation, and rostral/caudal ratio. Three of the characteristics showed a reliable decrease with increasing kinship with Old World monkey: surface density, volume density, and concentration. The reliable decrease in surface density is especially noteworthy because body weight was found to be a reliable predictor of surface density, accounting for 54% of its variation (Table 7). The 14 remaining characteristics showed no reliable change.

When brain weight (instead of body weight) was held statistically constant, only one characteristic, condensation, remained reliably and positively related to kinship with Old World monkey (Table 10). One additional charac-

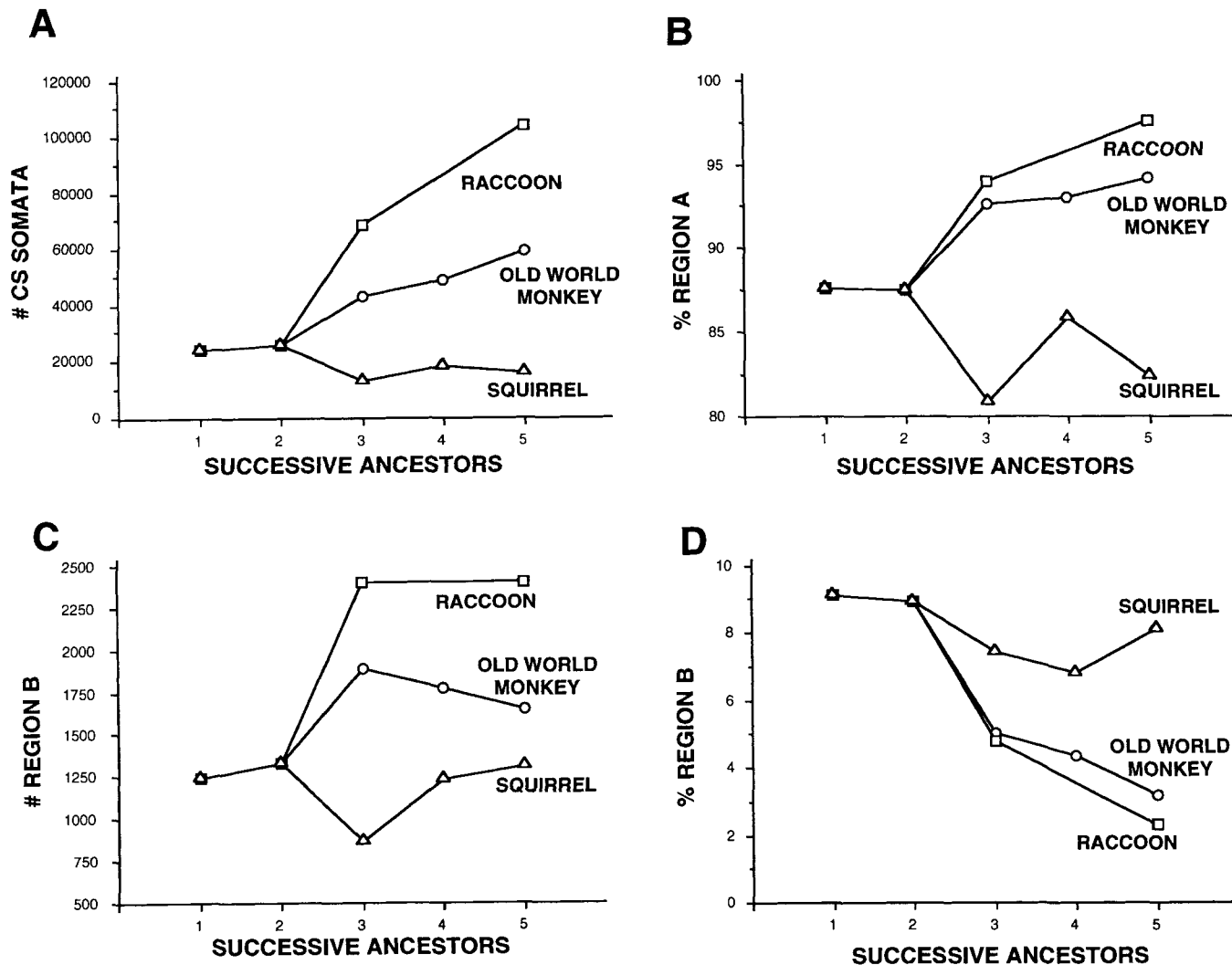


Fig. 7. A–D: Graphs show evolutionary changes in CS somata (total number of CS somata, percentage in region A, number in region B, and percentage in region B) in hypothetical ancestors of Old World monkey, raccoon, and gray squirrel. The graphs in Figures 7–9 show the CS characteristic averaged over the entire set of progeny of each hypothetical ancestor (or branch point) along each lineage.

teristic, concentration, was negatively related, i.e., the concentration of CS somata decreased with increasing ancestral closeness with Old World monkey.

Finally, when neocortical surface area was held statistically constant, only one relationship remained reliable. Concentration remained negatively related to kinship with Old World monkey, accounting for 25% of the variance (Table 11). Although concentration was not reliably related to any of the size parameters in the sample of mammals, stepwise regression revealed that the “best” size predictor of concentration was neocortical surface area.

The results suggest that most of the CS characteristics are more closely related to scaling factors than to kinship with Old World monkey. For example, the number of CS somata, their diameter, and their volume density are more closely related to neocortical surface area than to kinship with Old World monkey. Only the concentration of the CS somata themselves remained linearly related to kinship with Old World monkey alone, i.e., to kinship independent of either body, brain, or cortex size.

Relations of morphological characteristics of CS somata to the raccoon lineage that are independent of changes in body, brain, and neocortical size. When body size was statistically held constant, one characteristic, condensation, was positively related to kinship with raccoon (Table 9). When brain size or neocortex size were statistically held constant, condensation remained statistically reliable (Tables 10, 11). Neocortical surface area was the “best” predictor of condensation.

Relations of morphological characteristics of CS somata to the gray squirrel lineage that are independent of body, brain, and cortical size. When body size was held statistically constant, one of the 23 characteristics, thickness, was positively related to kinship with gray squirrel. One other characteristic, medial-lateral position, was negatively related to kinship with gray squirrel (Table 9).

When brain size was held statistically constant, thickness remained positively related, and medial-lateral position remained negatively related to kinship with gray squirrel

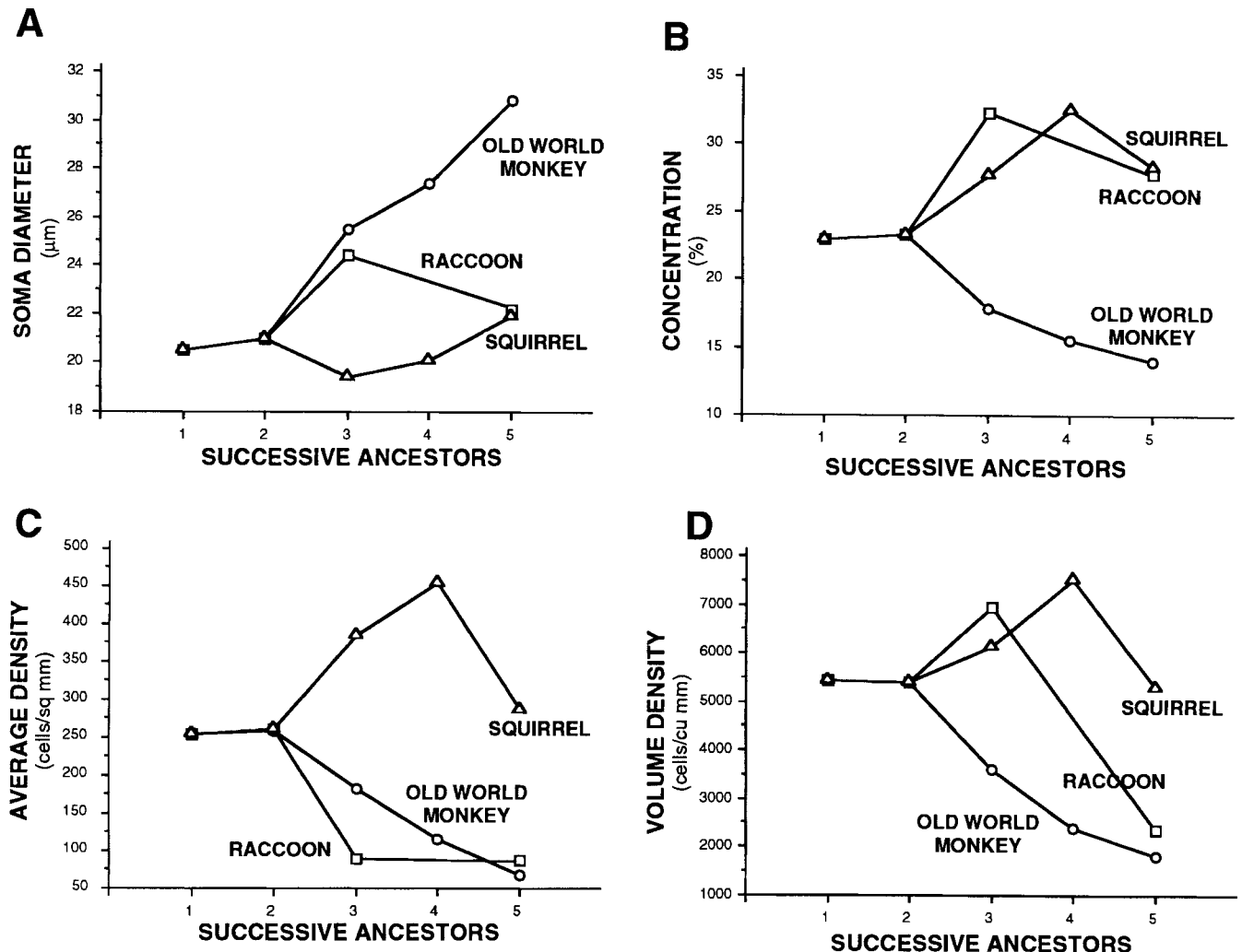


Fig. 8. A–D: Graphs show evolutionary changes in CS somata (soma diameter, concentration, average density, and volume density) in hypothetical ancestors of Old World monkey, raccoon, and gray squirrel.

(Table 10). Brain size was the “best” predictor of medial-lateral position.

Finally, when neocortex size was held statistically constant, four characteristics were positively related to kinship with gray squirrel: number of CS somata, soma diameter, thickness, and concentration of large somata (Table 11). Neocortex size was the “best” predictor of each of these four characteristics.

Summary of relations of CS somata characteristics to ancestry. In summary, the results show that 11 morphological characteristics of CS somata are linearly related to nearness of kinship with Old World monkey. Seven characteristics, including the number of CS somata, their soma diameter, thickness, condensation, and rostral/caudal ratio, are positively related to kinship with Old World monkey; four other characteristics, including the percentage of CS somata in region B, average surface area density, volume density, and concentration, are negatively related to kinship. Comparing these changes in the Old World monkey lineage with two other lineages, one leading to raccoon and the other leading to gray squirrel, shows that changes in six of the CS characteristics, soma diameter, rostral/caudal ratio, percentage of CS somata in region B, surface

density, volume density, and concentration, are reliable only in the Old World monkey ancestral lineage.

Finally, the results show that at least some of these changes in the Old World monkey ancestral lineage were probably not due to allometric, or scaling, factors or size changes. Nine of the 23 characteristics were linearly related to kinship with Old World monkey when body weight was mathematically held constant. One of these nine characteristics, surface density, was best predicted by body weight. Two characteristics, condensation and concentration, remained reliable when brain weight was held constant. Condensation was best predicted by brain weight as opposed to concentration, which was unrelated to brain weight (or to body weight or neocortical surface area). Only one characteristic, concentration of CS somata, uniquely remained reliable in the Old World monkey lineage when neocortical surface area was held constant.

DISCUSSION

In summary, the results show that the variation in several morphological characteristics of corticospinal somata is reliably related to variation in the size of an

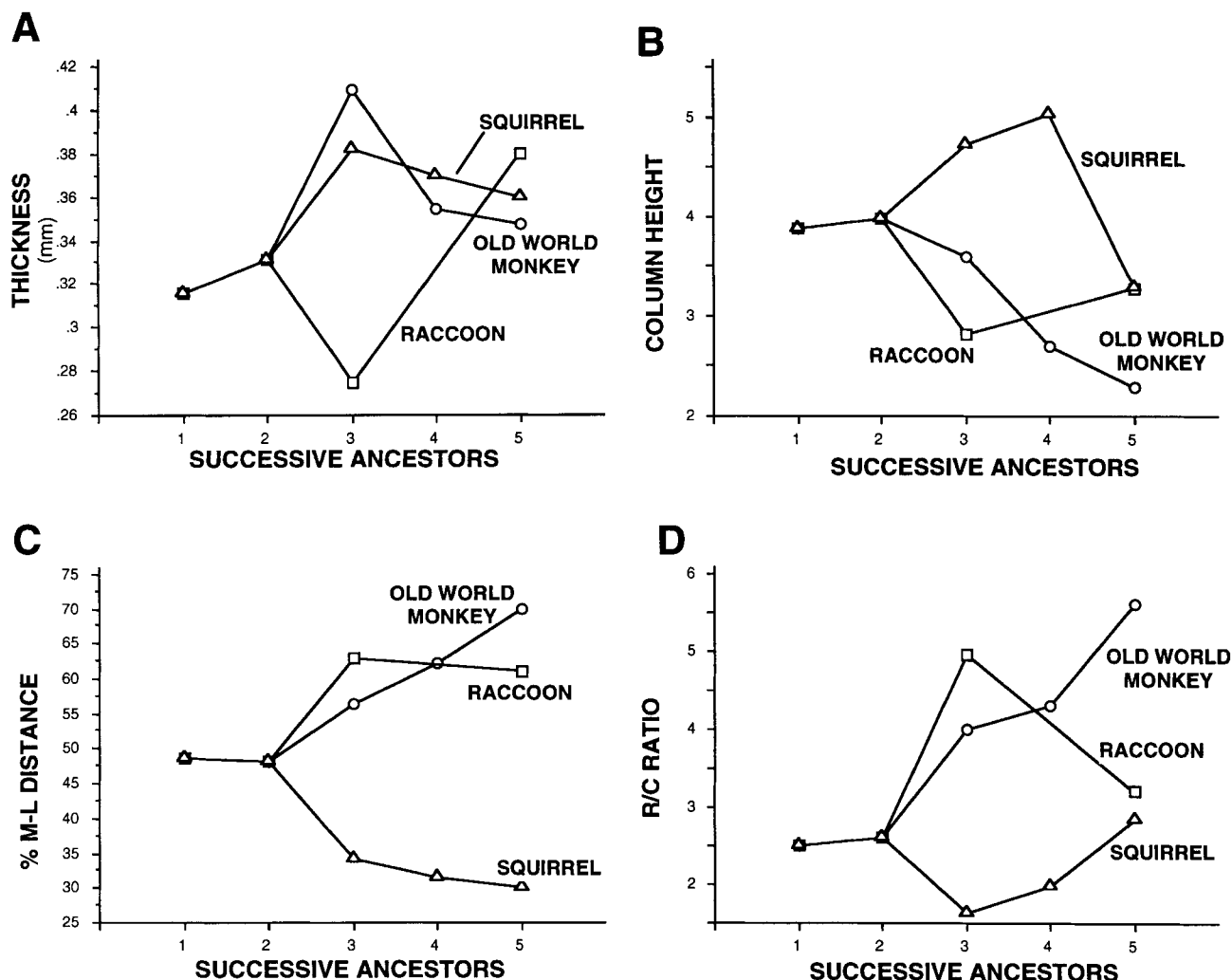


Fig. 9. **A–D:** Graphs show evolutionary changes in CS somata (thickness, column height, medial-lateral distance, and rostral/caudal ratio) in hypothetical ancestors of Old World monkey, raccoon, and gray squirrel.

animal's body, brain, or neocortical surface area, and part of this variation can be accounted for by differences between the primates and nonprimates in the sample. In addition, the results show that the variation in the morphology of CS somata is related to an animal's ancestry and that some of these relationships were not due simply to size changes. However, before discussing the possible functional significance of these relationships, a brief discussion of measurement and statistical error is indicated.

Sources of measurement error

The reliability of each of the statistical relationships ultimately depends on the reliability of the technique for estimating the numbers of CS somata. The possibility of under- and overlabeling errors, the extent to which these errors may have affected the results, and the appropriateness of the sample have been discussed in detail elsewhere (Nudo and Masterton, 1988, 1989, 1990a,b; Nudo et al., 1993). Briefly, because of the histological screening procedures employed, it is more likely than not that the present materials err toward underlabeling rather than overlabel-

ing the CST. That is, our procedures were designed more to eliminate "overlabeling" by inadvertent labeling of cells that are not part of the CST (such as corticobulbar fibers) than "underlabeling" by the nonlabeling of true CST fibers either by nonexposure to HRP or possibly by an HRP phobia of a subset of CST fibers.

In other words, we purposely eliminated all individuals in which the hemisection might have stopped short of the midline. These individuals typically had fewer labeled somata in the resulting material. Even in those individuals that received apparently complete hemisections, there was a possibility that the application of tracer was incomplete. Because our goal was to backfill the somata originating the CST in their entirety, and thus estimate the "true" value of CS somata, we selected the individual with the largest profile count as representing the species instead of averaging the individuals within each species. This bias may provide an overestimate of the true number of CS somata. Nevertheless, because the variation among species was much larger than the variation among individuals within a species, the results of statistical analyses were largely

TABLE 9. Relation of Corticospinal (CS) Somata to Three Ancestral Lineages Independent of Body Size

Characteristic	r Values for kinship with		
	Old World monkey	Raccoon	Gray squirrel
#CS somata	0.556**	0.406	0.290
#A	0.561**	0.401	0.268
#B	0.431*	0.399	0.273
#C	0.640		
#C'			0.217
%A	0.307	0.067	-0.259
%B	-0.410	-0.167	-0.136
%C	0.023		
%C'			-0.574
Density A	-0.239	0.215	0.365
Density B	-0.290	-0.172	0.232
Density C	0.359		
Density C'			0.400
Surface density	-0.412*	-0.072	0.288
Soma diameter	0.701**	0.168	0.306
Layer thickness	0.370	0.194	0.478*
Volume density	-0.465*	0.085	0.043
Column height	-0.182	0.153	0.244
Concentration	-0.465*	0.293	0.336
Concentration (large somata)	0.314	0.338	0.425
Condensation	0.562**	0.598**	0.322
Medial-lateral position	0.271	0.046	-0.444*
Rostral/caudal density ratio	0.507*	0.300	0.057

* $P < 0.05$; ** $P < 0.01$.

TABLE 11. Relation of Corticospinal (CS) Somata to Three Ancestral Lineages Independent of Total Neocortical Surface Area

Characteristic	r Values for kinship with		
	Old World monkey	Raccoon	Gray squirrel
#CS somata	0.011	0.277	0.417*
#A	0.000	0.269	0.386
#B	0.026	0.278	0.342
#C	0.362		
#C'			0.413
%A	-0.112	-0.064	-0.304
%B	0.014	-0.016	-0.128
%C	-0.272		
%C'			-0.479
Density A	-0.107	0.240	0.379
Density B	-0.114	-0.117	0.253
Density C	0.556		
Density C'			0.681
Surface density	0.086	0.064	0.333
Soma diameter	0.043	-0.147	0.442*
Layer thickness	0.135	0.086	0.513*
Volume density	-0.188	0.197	0.076
Column height	0.009	0.199	0.265
Concentration	-0.503*	0.358	0.333
Concentration (large somata)	0.006	0.216	0.497*
Condensation	0.405	0.510*	0.381
Medial-lateral position	-0.008	-0.077	-0.481*
Rostral/caudal density ratio	0.092	0.124	0.090

* $P < 0.05$.

TABLE 10. Relation of Corticospinal (CS) Somata to Three Ancestral Lineages Independent of Whole Brain Size

Characteristic	r Values for kinship with		
	Old World monkey	Raccoon	Gray squirrel
#CS somata	0.235	0.347	0.364
#A	0.232	0.341	0.336
#B	0.185	0.331	0.312
#C	0.322		
#C'			0.382
%A	-0.007	-0.023	-0.311
%B	-0.107	-0.069	-0.124
%C	-0.319		
%C'			-0.478
Density A	-0.112	0.234	0.384
Density B	-0.164	-0.142	0.254
Density C	0.440		
Density C'			0.658
Surface density	-0.014	0.027	0.354
Soma diameter	0.317	-0.009	0.403
Layer thickness	0.210	0.125	0.499*
Volume density	-0.244	0.165	0.079
Column height	-0.027	0.183	0.268
Concentration	-0.494*	0.331	0.335
Concentration (large somata)	0.144	0.266	0.458
Condensation	0.501*	0.549**	0.348
Medial-lateral position	0.079	-0.035	-0.479*
Rostral/caudal density ratio	0.278	0.204	0.068

* $P < 0.05$; ** $P < 0.01$.

unchanged when species averages were used instead of species maxima (i.e., for analyses using numbers of CS somata).

For the present purposes, a small bias toward larger values may be present in the measurements of soma diameter. Consequently, the number and density of labeled neurons as reported here were slightly biased as well. The average soma diameter used both as raw data and in the stereological correction for the total number of CS somata was determined by measuring random labeled cells within layer V of the section with the highest density (i.e., within region A). Although not specifically measured in every zone of every case, we noted that the average size of CS somata varied from region to region and among various zones within regions. The largest CS somata were invariably in

the zone of highest density within region A. Consequently, the reported "average" soma diameter is probably an overestimation of the mean diameter of all CS somata within a given case. The soma diameter was used as a correction factor for overcounting, therefore the numbers and the density of CS neurons are slightly underestimated.

Furthermore, the variability of soma diameter among regions and among zones within regions is lower for neurologically more generalized species (such as marsupials and insectivores) and higher for neurologically more specialized species (such as anthropoids and carnivores). This means that the bias in the corrected number of CS neurons in mammals with distant ancestry with anthropoids, such as marsupials, is less than in animals with more recent common ancestry with anthropoids, such as Old World monkeys.

To estimate the degree of this bias at its extreme, the soma diameter obtained in the highest density zone was compared with the soma diameter of a zone of considerably lower density (in the caudal portion of region A). The soma diameters obtained from the two regions differ by less than 10%. Furthermore, when the counts of CS somata are then corrected by using this lower value for the average soma diameter, the difference in the number of CS neurons is less than 3% of the number obtained by our standard procedure. Because the difference in the counts obtained by utilizing the two widely different soma diameters is small in this extreme condition, and because the measured cells actually used for correction were randomly selected within the zone of high density, the bias due to errors in mean soma diameter can be conservatively estimated as less than 3%.

A second small bias probably occurred in our measure of concentration. Because the concentration was determined only in the zone of peak density, the maximum concentration of CS somata in the entire case may be somewhat different. When lower density zones are examined, it appears that the concentration also is lower. Therefore, the measures of concentration reported here are probably reliable estimates of maximum concentration but are a slight overestimate of average concentration.

Finally, estimates of the density of CS somata are problematic along the banks of sulci in highly convoluted species (e.g., cat, raccoon, macaque, etc.). For this reason, both coronally and parasagittally sectioned material were used to validate our results. Although discrepancies were usually small, the supplemental material was necessarily derived from a different individual.

Adequacy of sample and statistical analyses

The present study was designed to examine the variation in form of the CS somata among mammals sufficient to derive the evolutionary trends in CS morphology in the anthropoid primate ancestral lineage. The best way to attain this goal (short of studying every species) is with a random sample of species whose ancestral histories are certain. However, few random samples of adequate size can be attained for practical reasons, i.e., almost all would fail to contain either rare or otherwise unavailable (e.g., protected) species. Therefore, although we examined 24 diverse species, they do not constitute a random sample. Even some major groups, for example, Chiroptera and Artiodactyla, are not represented, and others, for example, Rodentia, are underrepresented. Several representatives of each of these large orders would be expected in a truly random sample of 24 species of mammals. Therefore, generalizations to the entire class of Mammalia, even of the several characteristics found common to each of the 24 species, embodies extra statistical risk. Furthermore, the mammalian species in the sample (and any other sample) are each highly specialized and in many ways probably do not resemble the extinct ancestors. Neurological characteristics of extant mammals cannot be considered to be representative of those of extinct mammals. However, because we chose the animals in the present sample based, for the most part, on paleontological, taxonomic, and comparative anatomical grounds for their sequential kinship with anthropoid primates (as well as their availability, body size, cost, and other practical considerations), statistical calculation shows that the sample is a good one for this narrower evolutionary goal for which it was intended and at least an adequate one for somewhat wider comparative purposes, i.e., to make inferences concerning trends in human evolution. Further discussion of the adequacy of this sample for inferring homology can be found in a previous report (Nudo and Masterton, 1988).

Furthermore, the best way to examine relationships between the variation in morphology and the anthropoid ancestral lineage is to perform correlation and regression analyses by using the actual dates of branch points in mammalian evolution. Although it is possible in principle to derive dates of divergence based on fossil evidence, the likelihood of underestimation is high (Martin, 1990). Therefore, we have employed an ordinal scale that ranks the species in the sample according to successive kinship with Old World monkeys, raccoon, or gray squirrel. To the extent that linear correlation and regression analysis is more applicable to interval data than to ordinal data, the present conclusions must be weakened accordingly. However, the results of simple linear correlation, partial correlation, and regression analyses reported here are at least useful for describing general trends in CS morphology along the three lineages.

It should also be noted that new molecular and morphological data suggest that New World edentates, which include the armadillo, may represent an early diverging

branch of placental mammals (Novacek, 1992). In the present statistical treatment, we have conservatively included armadillos with the other Eutherian nonprimates. However, if an additional grade is added to our scale, (i.e., with armadillos intermediate between marsupials and other Eutherian nonprimates), the results are essentially unchanged.

Finally, the individuals selected for study were not matched for sex or age, although all were at least young adults. To the extent that sex or age may contribute to the variability in the number, size, or density of CS somata, conclusions based on these results should be weakened accordingly. However, an attempt to trace possible morphological changes in the somata originating the CST based on these same animals seems justified if only on the basis of it being the best sample now available.

Comparisons with previous reports of CS neurons in mammals

The present results are in general agreement with previously published descriptions with respect to the distribution of CS neurons in those species that have been reported (Hicks and D'Amato, 1977; Toyoshima and Sakai, 1982; Hutchins et al., 1988; Li et al., 1990; Sakai, 1990; Dum and Strick, 1991; Holst et al., 1991; He et al., 1993). However, because virtually all previous studies of the neurons originating the CST relied on injections of retrograde tracer into the spinal cord, a direct comparison of the total number of CS somata in various mammalian species has not been possible. Indeed, the stereologically corrected numbers presented here are higher than any reported for the same species; the hemisection, bulk-filling method used here probably approaches the totality of CST filling, whereas no such claim can be, or has been made, when using injections of tracer regardless of their size or multiplicity. Therefore, within the constraints of sampling errors discussed earlier, our standardized bulk-filling technique for maximal labeling of CS neurons would seem to provide a closer approximation of the total number of CS somata across the sample of mammalian species.

Relationships between morphological characteristics of CS somata and body size, brain size, and neocortical surface area

Several morphological characteristics of CS somata (total number of CS somata, number and percentage in region A, soma diameter, etc.) are positively related to body, brain, or neocortex size. For simplification, the following discussion is limited primarily to relationships with "best" predictors using stepwise regression analysis. For example, neocortical surface area was found to be the best predictor of 11 CS characteristics. Variation in surface area accounted for 69% of the variation in CS number and 82% of the variation in CS soma diameter. Such positive relationships suggest that these CS characteristics are related to size according to strictly allometric, or scaling, functions but more so for neocortex area than for brain weight or body weight. These results seem to indicate that the number and size of CS somata are not entirely predicted on the basis of an animal's body size or brain size alone. The number and size of CS somata are more closely predicted by the size of an animal's neocortical surface, including both CS and non-CS areas.

In sharp contrast, other morphological characteristics of CS somata (e.g., density in region A, average surface

density, column height, etc.) are negatively related to body, brain, or neocortex size. All but one of the negative relationships with size resulted from derived characteristics based on ratios of two direct measurements. These relationships may be negative even though both the numerator and denominator are positively related to size because the denominator may increase at a faster rate. For example, the percentage of CS somata in region B is negatively, but reliably, related to body, brain, and neocortex size. Both the numerator (number of CS somata in region B) and the denominator (total number of CS somata) are positively related to brain and neocortex size. However, the slope of the linear regression line for the total number of CS somata is steeper than for the number of CS somata in region B, implying that with increased neocortex size, CS somata are added to region B at a lesser rate than they are added to the total. Therefore, the negative relationships between the percentage of CS neurons in region B and neocortex size are due to a relatively smaller increase in the number of CS somata in region B compared with the increase in total number of CS somata.

Likewise, surface density is negatively, but reliably, related to body size. However, both the numerator (total number of CS somata) and the denominator (total neocortical surface area containing CS somata) increase with increased size (present results; see Nudo and Masterton, 1990b). Apparently, with increased size the amount of CS cortex increased faster than the total number of CS somata. Therefore, these derived characteristics are negatively related to size due to the consequences of arithmetic operations.

However, some relationships based on derived characteristics may be negative because the numerator alone is negatively related to size. For example, the maximum density of CS somata in regions A, B, and C are negatively, but reliably, related to body size. Because the denominator for these measurements is a sample area of constant size, the negative relationships are due to decreases in the numerator, i.e., the number of CS neurons subjacent to the sample area. Therefore, these negative relationships are not due to the consequences of arithmetic operations.

As another example, column height was directly measured from the histological material in absolute numbers. Therefore, the negative relationship between column height and body size cannot be due to the consequences of arithmetic operations.

Finally, maximum volume density was negatively, but reliably, related to neocortex size. Once again, because volume density is derived by dividing the surface density by the thickness of the CS somata layer, and because thickness is positively and reliably related to neocortex size, the negative and reliable relationships between volume density and neocortex size are probably due both to decreases in the number of CS somata in the sample volume and to increases in the thickness of the labeled layer.

Therefore, it would appear that several morphological characteristics of CS somata vary due to different body, brain, and neocortical sizes. In general, mammals with large neocortical surface areas tend to have more and larger CS somata, and these somata are less densely distributed within layer V. Taken together with previous findings that the total neocortical surface area devoted to CS somata is positively related to body size and cortex size (Nudo and Masterton, 1990b), it seems that mammals with large bodies and large brains achieved an increase in the number

of CS somata by a substantial increase in neocortical surface area and a concomitant decrease in CS density. These relationships are most marked in region A (i.e., regression coefficients are higher in region A than in region B), the region that encompasses primary motor and somatosensory cortex, and, at least in primates, premotor and supplementary motor areas.

Because of the close relationship between brain size and neocortex size, most of the relationships between CS characteristics and brain size are similar to those between CS characteristics and neocortex size. However, the relationships that differ are noteworthy because they may provide insight into the variation in cortical versus noncortical brain regions and how these two sources of variation are related to CS characteristics. For example, the number and size of CS somata, the percentage of CS somata in region A, the thickness of the layer of CS somata in layer V, and the volume density are each related more closely to cortical surface area than to brain size (Table 6). These relationships may reflect primarily cortical sources of variation. That is, the changes in these CS characteristics with neocortical size appear to be unrelated to increases in the size of noncortical brain regions.

In sharp contrast, two CS characteristics, number of CS somata in region C among primates and medial-lateral position of highest density, are more closely related to brain size than to neocortex size. Thus, these relationships may reflect strictly noncortical sources of variation, i.e., the decrease in the number of CS somata in region C and the more lateral position of CS somata with increased brain size each appear to be unrelated to increased size of neocortex.

Finally, although allometric scaling is an important contributor to the variation in morphology of corticospinal somata, it is a relatively weak contributor to the variation in morphology of somata originating other descending spinal pathways. For example, the number and size of tectospinal somata are not reliably related to body weight, brain weight, tectal volume, or tectal surface area (Nudo and Masterton, 1989). Thus, allometric scaling may be a larger factor in those descending spinal pathways that extend to more caudal segments of the cord.

Divergence of allometric trends in primates

Although many of the allometric relationships between the 23 CS characteristics and the size of the body, brain, and neocortex are similar in primates and nonprimates, others are not. Because these differences may reflect important adaptations or possibly unique evolutionary divergence in primates, each warrants special consideration.

First, the medial-lateral position of the zone of highest CS density within region A is positively and reliably related to brain size in primates as opposed to nonprimates. Therefore, the trend toward a more lateral position for the high density zone may be related to allometric scaling but only in primates. Because high-density zones are typically in the rostral, or motor, portion of region A, and because more lateral zones probably correspond to functional representations of the hand, it would appear that primates with larger brains possess relatively more CS somata concerned with motor control of the hand than smaller primates. The position of the zone of highest CS density in bushbaby is the most medial of any of the species in the sample. This more medial position probably corresponds to the hindlimb or tail representation rather than the forelimb. Therefore, conversely, it may be said that primates with smaller brains

possess relatively more CS neurons concerned with motor control of the hindlimb and/or tail than larger primates. Alternatively, the more medial position of CS neurons (especially in bushbaby) may be a specialization associated with large, powerful hindlimbs.

Second, three characteristics, density in region A, density in region B, and column height, are negatively, but reliably, related to body, brain, and cortex size in primates but not in nonprimates. This result would seem to indicate that a lower density of CS somata in primates is the result of unique scaling relationships, possibly reflecting a unique organization of the neocortex itself.

In addition, some characteristics are reliably related to body, brain, or neocortex size in both primates and nonprimates, but their linear regression lines display significantly different slopes or elevations. First, surface density was negatively, but reliably, related to body weight in both primates and nonprimates but decreased at a considerably faster rate (over two times faster) in primates (Fig. 4). Second, primates have more and larger CS somata than mere body weight suggests but not more than neocortical area suggests (Fig. 3). Therefore, although the number and size of CS somata are related to body weight by similar functions in primates and nonprimates, primates possess more and larger CS somata for any given body size.

Evolutionary changes in anthropoid ancestry

The results indicate that once the CST emerged in ancient mammals (i.e., in the common ancestor of the mammals in the sample), several further changes took place in the morphology of the cortical somata originating the tract. As discussed in previous papers, (Nudo and Masterton, 1988, 1990a,b), at least two entirely new cortical regions appeared after the chief divergence of placental mammals (region C found in primates and region C' found in rodents and in rabbit).

In addition, the present results suggest that several quantitative changes took place. Statistically reliable changes in the number, size, and density of CS somata certainly occurred in the ancestry of Old World monkeys, raccoon and gray squirrel, and probably in the ancestry of the other animals as well.

The present results show that several changes probably took place in the morphology of the CS somata in the anthropoid ancestral lineage. Increasing recency of common ancestry, or nearness of kinship, with Old World monkeys is reliably related to increases in the total number of CS somata, their diameter, and the thickness of the layer of CS somata within layer V. Furthermore, condensation, which is a relative measure of heterogeneity in the distribution of CS somata, and the ratio of their rostral/caudal density probably increased. These results indicate that in the anthropoid ancestral lineage, CS somata became relatively more dense in motor (rostral) areas of the cortex.

Perhaps unexpectedly, recency of common ancestry with Old World monkeys is also reliably related to decreases in surface density, volume density, and concentration. These results mean that both the area and volume of cortex associated with each CS neuron probably increased to accommodate local non-CS "server" cells.

In general, the evolution of the CST in the anthropoid primate lineage was marked by increase in total numbers of somata, increase in soma size, decrease in density, and substantial changes in their distribution within the neocortex.

As already described, although these trends in the evolution of the CST may have been important in their own right, many of these same characteristics also covary closely with parallel increases in body, brain, and neocortical size. Because characteristics that covary with body size may be related to simple allometric or scaling factors (e.g., differences in skeletal muscle mass, body length, limb length, etc.), the analysis of each of the characteristics holding body size statistically constant are a necessary part of uncovering which characteristics are more likely to be of special functional significance.

The present analyses show that several morphological characteristics of CS somata probably changed reliably in anthropoid ancestry, without simply mirroring concurrent changes in body size. For example, in the anthropoid primate lineage, when body weight is held statistically constant, the increase in the total number of CS somata and the number of CS somata in each of regions A and B, their soma diameter, their condensation, and their rostral/caudal ratio remain reliable. Likewise, when body weight is statistically held constant, the decrease in surface density, volume density, and concentration remain reliable. Finally, when body weight is statistically held constant, thickness of the layer of CS somata increased and then decreased in pre- and postprosimian stages respectively.

These results suggest that at least the changes in number, size, and density of CS somata in anthropoid ancestry are probably related to changes in neural organization and not simply to the allometric changes related to the neural support of a larger body mass.

For a given body weight, brain size is considerably larger in primates versus most other mammals (e.g., Jerison, 1973; Stephan et al., 1981; Hofman, 1985). Therefore, CS characteristics that changed reliably in anthropoid ancestry independent of body size still may be related to increases in relative brain size. Because characteristics that covary with brain size may be related to neural changes in both cortical and noncortical brain structures, the analysis of each of the characteristics holding brain size statistically constant may reveal changes related to neocortex alone. For example, the number of CS somata, their diameter, and their density are not reliably related to kinship with anthropoid primates when brain weight is held statistically constant. Therefore, changes in these characteristics are somehow related to a relative increase in brain size. This result suggests that although significant increases in the number and size of CS somata took place during anthropoid ancestry, and although these changes were not simply related to a larger body mass, the entire brain (as well as the entire neocortex and the part of cortex occupied by the CS somata) grew at similar rates. We have shown elsewhere that the percentage of CS cortex relative to total neocortex remained almost constant in the human lineage (Nudo and Masterton, 1990b), so it follows that the increase in the number of CS somata did not keep pace with either the expansion of the CS cortex itself or with the total area of neocortex. However, it is also true that these same trends illustrate that there was an increase in the total amount of CS neocortex per CST axon.

In contrast, some CS characteristics probably changed reliably in anthropoid ancestry independent of whole brain size. For example, when brain size was held statistically constant, condensation increased reliably and concentration decreased reliably with increasing kinship with anthropoid primates. This means that changes in condensation

and concentration were greater than expected on the basis of changes in brain size alone. Therefore, these changes may be related to neural changes within the neocortex itself.

In this regard it is notable that concentration remains reliably related to kinship with anthropoid primates when neocortical surface area is held statistically constant. This result suggests that the changes in the concentration of CS somata in anthropoid ancestry were probably related to changes in the architecture of the CST itself and independent of changes in neocortical surface area.

Evolutionary changes in raccoon's and gray squirrel's ancestry in comparison with changes in anthropoid ancestry

Some of the same characteristics that are reliably related to the Old World monkey ancestral lineage are also reliably related to raccoon's ancestral lineage, illustrating that these changes were not unique to anthropoid ancestry. From a functional perspective, this means that these changes may underlie the commonalities of raccoons and anthropoid primates and their differences from other mammals, but they probably do not underlie any differences between anthropoid primates and raccoons. For example, the total number of CS somata, the number in regions A and B, and their condensation increased in both lineages.

However, comparing changes among the three lineages (Old World monkeys, raccoon, gray squirrel) reveals that other changes in CS characteristics may indeed have been unique to the anthropoid primate lineage (and thus presumably unique to the human lineage). Chief among these possibly unique characteristics of the anthropoid lineage are the increase in CS soma diameter and the rostralward shift in the peak density of CS somata. These two characteristics may have increased only in the anthropoid lineage. Thus, CS somata probably became larger in anthropoid ancestry (i.e., larger than predicted based on scaling factors alone), and CS neuronal density in the more motor areas of cortex probably increased relative to CS neuronal density in the more sensory areas of cortex.

Finally, in addition to these possibly unique increases in the anthropoid lineage, CS concentration and volume density decreased only in the anthropoid lineage of the three lineages analyzed. Thus, the ratio of non-CS to CS somata within the CS areas probably increased, and this increase may have been unique in the ancestral lineage leading to anthropoid primates. Taken together with the result that the number of CS somata did not keep pace with the growth of the CS cortex itself, and that the CS cortex did keep nearly constant pace with the growth of the entire neocortex, it must have done so by increasing the number of non-CS somata within the CS cortex and not increasing the number of CS somata themselves.

One other CS characteristic that possibly changed uniquely in anthropoid ancestry deserves consideration. The percentage of CS somata in region B probably decreased only in the anthropoid primate ancestry. Because region B corresponds to the second somatosensory area (or SII), these results suggest that the functional importance of CS somata in SII may have progressively declined in anthropoid ancestry relative to other cortical areas originating CS fibers, such as premotor, supplementary motor, and primary motor cortex.

Comparison of changes in CS soma morphology with other evolutionary changes in corticospinal tract morphology

In a review of evolutionary changes in the fibers contained in the pyramidal tract, Heffner and Masterton (1983) found that the overall size of the tract probably increased throughout primate evolution. This increase was shown to be independent of parallel increases in body weight. However, it was also shown that mere ancestry is not reliably related to either the size or the total number of the fibers in the tract. In the present study, both the total number of CS somata and their diameters reliably increase in the anthropoid lineage, supporting this idea of an enlarging tract penetrating farther down the cord. Moreover, because the relationship between soma diameter and kinship with anthropoid primates remains reliable even when body weight is statistically held constant, it can be suggested that these changes in CS neurons in the pyramidal tract probably were not merely scaling changes necessary to support an enlarging body.

It is possible that the lack of reliable relationships between number of pyramidal fibers and fiber size in the study by Heffner and Masterton (1983) may have been due to the smaller sample of mammals for which these data were available. However, it is also possible that the discrepancy between the results of these two studies may indicate an important difference between the corticospinal tract and the pyramidal tract. For example, because many pyramidal tract fibers terminate in the lower medulla before reaching the spinal cord, the two tracts are not identical. Nevertheless, for the 12 species examined in the present study for which pyramidal tract data is also available, over 93% of the variance in pyramidal tract area ($r = 0.965$, $r^2 = 0.93$) can be accounted for by two CS neuron characteristics: the total number of CS somata and (the square of) their diameters. Thus, the high and reliable relationship between area of the pyramidal tract and kinship with humans is probably due simply to the closely parallel increases in both the size and the number of corticospinal somata.

In summary, the evolution of the CST in the anthropoid primate lineage (and presumably in the human lineage) was marked by increases in total numbers of somata, increases in soma size, decreases in density, and substantial changes in their distribution within the neocortex. After comparing these changes with those changes in the lineages of raccoon and squirrel, the results suggest that if morphological characteristics of the CST were important in the evolution of specialized motor abilities in anthropoid primates, then examination of the role of changes in soma diameter, rostral (motor)/caudal (sensory) ratio, concentration, surface density, and volume density may be more instructive than examination of total number of CS neurons alone.

These differences between primates and nonprimates suggest that several important, if not unique, changes probably occurred in the anthropoid lineage alone. It is possible that these structural differences in CS somata may underlie changes in motor function in the anthropoid lineage. For example, it is tempting to guess that the large relative decrease in concentration and surface density of CS somata in primates may be related to an increase in local circuit neurons and in the cortical neuropil related to the CS neurons. A relative decrease in projection neurons, with a concomitant increase in local circuit neurons, may allow for a more adaptable system (Hofman, 1985). The special-

ized motor abilities of primates at least partially may be due to the development of a more highly adaptable cortical motor system rather than merely a larger one.

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